

smokers may not get quite the same results. What we care about is the mean for all smokers. It is impractical to measure that mean, but if we know the sample mean, the size of the sample, and the standard deviation, we can estimate whether the sample mean is likely to be close to the population mean. Figure 2.21 presents two possibilities. In part (a), the 95% confidence intervals are small; in other words, the standard deviations were small, the samples were large, and the sample means are almost certainly close to the true population means. In part (b), the confidence intervals are larger, so the numbers 7.5 and 6.5 are just rough approximations of the true population means. Presenting data with confidence intervals can enable readers to decide for themselves how large and impressive the difference is between two groups (Hunter, 1997; Loftus, 1996).

A 95% confidence interval is one kind of **inferential statistic**, which is a *statement about a large population based on an inference from a small sample*. One alternative to the 95% confidence interval is a test that determines the probability of achieving a difference as large as the observed one by purely chance variation. For example, if we compared two groups of smokers who were actually undergoing the same therapy, what would be the chance that the difference between the two groups would be at least as large as the difference between 7.5 and 6.5? The result is summarized by a p (as in *probability*) value. For example, $p < .05$ indicates that *the probability that randomly generated results would resemble the observed results is less than 5%*. The smaller the p value, the more impressive the results. The usual convention is that, if p is less than 0.05, researchers consider the results to be **statistically significant** or **statistically reliable**—that is, *unlikely to have*

arisen by chance. A more cautious researcher might consider the results to be significant or reliable only if p were less than 0.01 or even 0.001. The appendix at the end of this chapter gives an example of a statistical test that can be used to determine a p value. Statistical significance depends on three factors: the size of the difference between the groups, the number of research participants in each group, and the amount of variation among individuals within each group (Figure 2.22).

CONCEPT CHECK

- 12.** Should we be more impressed with results when the 95% confidence intervals are large or small? Should we be more impressed if the p value is large or small? (Check your answer on page 58.)

Examining the statistics is only the first step toward drawing a conclusion. To say that an experiment has statistically reliable results means only that we would be unlikely to get such results merely by chance. If not by chance, then how? At that point, psychologists use their knowledge to try to determine the most likely interpretation of the results.

THE MESSAGE Statistics and Drawing Conclusions

Sometimes psychological researchers get consistent, dependable effects, and they do not even have to consider statistics: Turn out the lights in a sealed room; people will no longer be able to see. Add sugar to your iced tea; it tastes sweet. The bigger the effect, the less we need to rely on complicated statistical tests. We pay more attention to statistics when we measure smaller effects: Does a change in the wording alter people's responses to a survey? Does the use of an electronic study guide improve students' test scores? Does family therapy provide better results than individual therapy to treat alcohol and drug abuse? Psychologists frequently deal with small, fragile effects and therefore need a solid understanding of statistics.

SUMMARY

- ★ **Mean, median, and mode.** One way of presenting the central score of a distribution is via the mean, determined by adding all the scores and dividing by the number of individuals. Another way is the median, which is the middle score, after all the scores have been arranged from highest to lowest. The mode is the score that occurs most frequently. (page 53)
- ★ **Standard deviation (SD).** To indicate whether most scores are clustered close to the mean or whether they are spread out, psychologists report the range of scores, called the standard deviation. If we know that a given score is a certain number of standard deviations above or below the

Factors that lead to a conclusion of “statistically significant” or “not statistically significant”

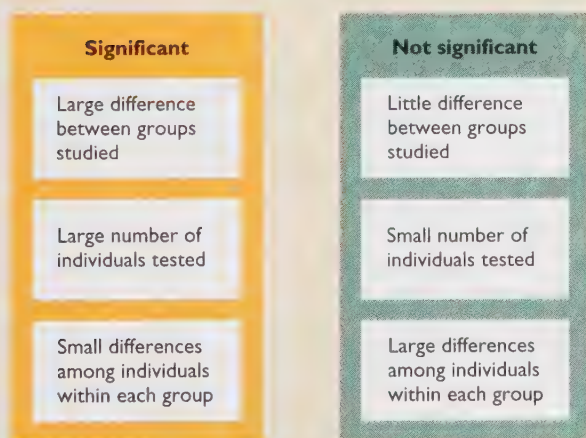


FIGURE 2.22 Researchers say that results are *statistically significant* if they calculate that chance variations in data would be unlikely to produce a difference between groups as large as the one that the researchers actually observed.

mean, then we can determine what percentage of other scores it exceeds. (page 55)

★ **Inferential statistics.** Inferential statistics are attempts to deduce the properties of a large population based on the results from a small sample of that population. (page 56)

★ **Probability of chance results.** The most common use of inferential statistics is to calculate the probability that a given research result could have arisen by chance. That probability is low if the difference between the two groups is large, if the variability within each group is small, and if the number of individuals in each group is large. (page 57)

★ **Statistical significance.** When psychologists say $p < .05$, they mean that the probability that accidental fluctuations could produce the kind of results they obtained is less than 5%. They generally set a standard of 5% or less. If the results meet that standard, they are then said to be statistically significant or reliable. (page 57)

Suggestion for Further Reading

Martin, D. (1996). *Doing psychology experiments* (4th ed.). Pacific Grove, CA: Brooks/Cole. A discussion of all aspects of research, including methods of conducting research and statistical analyses of results.

Terms

descriptive statistics mathematical summaries of results, such as measures of the average and the amount of variation (page 53)

mean the sum of all the scores reported in a study divided by the number of scores (page 53)

normal distribution (or **normal curve**) a symmetrical frequency of scores clustered around the mean (page 54)

median the middle score in a list of scores arranged from highest to lowest (page 54)

mode the score that occurs most frequently in a distribution of scores (page 54)

range a statement of the highest and lowest scores in a distribution of scores (page 55)

standard deviation (SD) a measurement of the amount of variation among scores in a normal distribution (page 55)

95% confidence interval the range within which the true population mean lies, with 95% certainty (page 56)

inferential statistics statements about large populations based on inferences from small samples (page 57)

$p < .05$ an expression meaning that the probability of accidentally achieving results similar to the reported results is less than 5% (page 57)

statistically significant or **statistically reliable results** effects that have a low probability of having arisen by chance (page 57)

Answers to Concept Checks

9. a. Mean = 4; median = 3; mode = 2. b. Mean = 7; median = 3; mode = 2. Note that changing just one number in the distribution from 8 to 35 greatly altered the mean without affecting the median or the mode. (page 54)
10. Most of the men replied "1," indicating a preference for a monogamous relationship. However, some specified much higher numbers (perhaps even in the thousands), raising the mean. (page 54)
11. Even though your score rose from 80 on the first test to 90 on the second, your performance actually deteriorated in comparison to other students' scores. A score of 80 on the first test was two standard deviations above the mean, better than 98% of all other students. A 90 on the second test was only one standard deviation above the mean, a score that beats only 84% of the other students. (page 56)
12. In both cases, smaller. A small 95% confidence interval indicates high confidence in the results. A small p value indicates a low probability of getting such a large difference merely by chance. (page 57)

Web Resources

Statistical Assessment Service (STATS)

www.stats.org/

Here you'll learn how statistical and quantitative information and research are represented (and misrepresented) by the media, and how journalists can learn to convey such material more accurately and effectively. The *Vital STATS* newsletter and STATS Spotlight include many interesting articles; Dubious Data Awards provide hilarious reading about the misuse of statistics; Newsclips are more serious examples of mangled and misunderstood data.

Statistical Calculations

This appendix shows you how to calculate a few of the statistics mentioned in Chapter 2. It is intended primarily to satisfy your curiosity. Ask your instructor whether you should use this appendix for any other purpose.

Standard Deviation

To determine the standard deviation (SD):

1. Determine the mean of the scores.
2. Subtract each of the individual scores from the mean.
3. Square each of those results, add the square roots together, and divide by the total number of scores.

The result is called the *variance*. The standard deviation is the square root of the variance. See Table 2.3 for examples.

A Typical Statistical Test: The *t*-test

Several statistical tests are available to suit different kinds of data and different kinds of experiments. For the simple purpose of comparing two groups, both of which show results that approximate the normal distribution, one of the most popular tests is the *t*-test. Assume that for two populations the means are $\bar{x}_1$ and $\bar{x}_2$, the numbers of individuals measured are n_1 and n_2 , and the standard deviations are s_1 and s_2 .

TABLE 2.3

INDIVIDUAL SCORES	MEAN MINUS THE INDIVIDUAL SCORES	DIFFERENCES SQUARED
12.5	2.5	6.25
17.0	-2.0	4.00
11.0	4.0	16.00
14.5	0.5	0.25
16.0	-1.0	1.00
16.5	-1.5	2.25
17.5	-2.5	6.25
		36.00

Mean = 15.0
 Variance = $36/7 = 5.143$
 Standard deviation = 2.268

We calculate t by this formula:

$$t = \frac{(\bar{x}_2 - \bar{x}_1) \sqrt{n_1 \cdot n_2 \cdot (n_1 + n_2 - 2)}}{\sqrt{n_1 \cdot s_1^2 + n_2 \cdot s_2^2} \cdot \sqrt{n_1 + n_2}}$$

The larger the value of t , the less likely that the difference between the two groups is due to chance. The value of t will be high if the difference between the two means ($\bar{x}_2 - \bar{x}_1$) is large, if the standard deviations (s_1 and s_2) are small relative to the means, and if the number of individuals is large. For example, if a group of 50 people has a mean of 81 and a standard deviation of 7, and a group of 150 people has a mean of 73 and a standard deviation of 9; then

$$t = \frac{(81 - 73) \sqrt{150 \cdot 50 \cdot 198}}{\sqrt{(150 \cdot 81 + 50 \cdot 49)} \sqrt{(200)}} = \frac{9748.8}{120.83 \times 14.14} = 5.71$$

The larger the value of t , the less likely that the results have arisen by accident. Statistics books contain tables to show the likelihood of a given t value. In this case, with 2 hundred people in the two groups combined, a t value of 5.71 is significant ($p < 0.001$).

Correlation Coefficients

To determine the correlation coefficient, we designate one of the variables x and the other one y . We obtain pairs of measures, x_i and y_i . Then we use the following formula:

$$r = \frac{[(\sum x_i y_i) - n \cdot \bar{x} \cdot \bar{y}]}{n \cdot s_x \cdot s_y}$$

In this formula, $(\sum x_i y_i)$ is the sum of the products of x and y . For each pair of observations (x, y) , we multiply x times y and then add together all the products. The term $n \cdot \bar{x} \cdot \bar{y}$ means n (the number of pairs) times the mean of x times the mean of y . The denominator, $n \cdot s_x \cdot s_y$ means n times the standard deviation of x times the standard deviation of y .

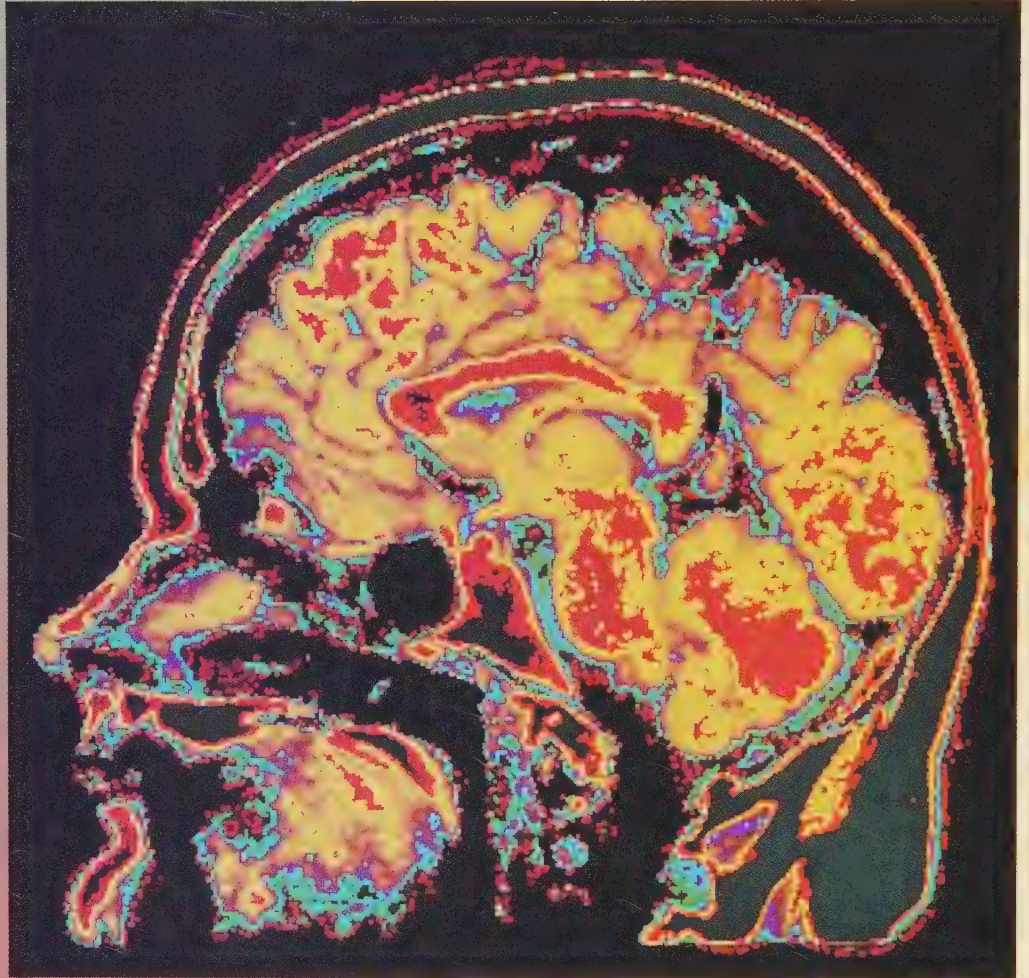
Web Resources

Introductory Statistics: Concepts, Models, and Applications

www.psychstat.smsu.edu/introbook/sbk00.htm

You can read an entire statistics textbook by David W. Stockburger, Southwest Missouri State University, on the Web or download it, free!

Biological Psychology



3

★ MODULE 3.1

Psychological Explanations at the Level of the Gene

Principles of Genetics

*Sex-Linked and Sex-Limited Genes / Identifying and
Localizing Genes / Measuring Heritability in Humans*

The Route from Genes to Behavior

How Heredity Interacts with Environment

Evolution

The Evolution of Behavior

Sociobiology

*Animal Examples of Sociobiological Explanations /
Speculations About Human Sociobiology*

★ MODULE 3.2

Explanations of Behavior at the Level of the Neuron

The Cells of the Nervous System

The Action Potential

Synapses: The Junctions Between Neurons

NEURONS COMMUNICATE BY RELEASING CHEMICALS: WHAT'S
THE EVIDENCE?

Neurotransmitters and Behavior: The Example of
Parkinson's Disease

★ MODULE 3.3

Explanations of Behavior at the Level of the Nervous System

A Survey of the Nervous System

*The Spinal Cord / The Autonomic Nervous System /
The Endocrine System / Between the Spinal Cord and
the Cerebral Cortex*

The Organization and Functioning of the Brain

*Effects of Localized Brain Damage: The Visual
System / Effects of Localized Brain Damage: Unilateral
Neglect / Effects of Localized Brain Damage: Specific
Memory Loss / Conclusions About Brain Organization /
Division of Labor: The Four Lobes of the Cerebral
Cortex*

Effects of Experience on Brain Structure

The Corpus Callosum and the Split-Brain Phenomenon

*Connections Between the Eyes and the Brain /
Behavioral Effects of Severing the Corpus Callosum*

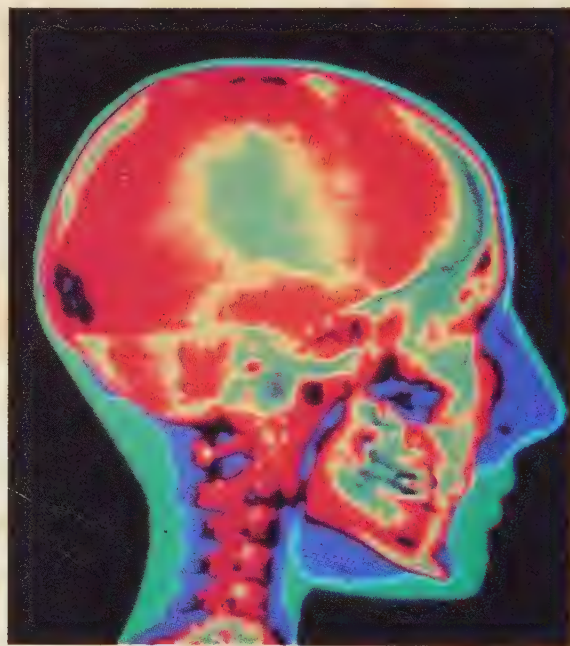
Common Misunderstandings About the Brain

It is easy to look at a human brain, which weighs only 1.2 to 1.4 kg (2 ½ to 3 pounds), and marvel at its amazing abilities. But let's contemplate some even more compact brains. A bee has a brain (if you want to call it that) so small that one needs a microscope to find it. And yet a bee's behavior is remarkably complex. A bee flies around until it locates flowers, finds the food in the flower, evades predators, finds its way back to the hive, and then does a dance that tells other bees where it found the food. When necessary, it also takes care of the queen bee, protects the hive against intruders, and so forth. The bee's very tiny brain can do some incredible tasks.

What does it feel like to be a bee? In fact, does it feel like anything, or does a bee have no more consciousness than an automobile? We don't know; we cannot get inside the brain of a bee to find out. In fact, you and I cannot get inside each other's heads, either. When we speak of "experiences like yours and mine," we are really just inferring each other's experiences. We do not know them directly.

Researchers necessarily proceed piecemeal, first answering the questions that are easiest to answer. We now know a great deal about how nerves work, how brains record sensory experiences, and how various kinds of brain damage interfere with behavior and experience. What we understand least is why brain activity produces experience at all. We have much evidence indicating genetic influences on behavior, but in most cases we still know little about how these genes exert their effects. Progress in brain research has been enormous, and yet we can be just as easily awed by how much we have learned or how much we do not yet know.

Will we someday understand nerves well enough to "read the mind" of a bee? or even to decide whether a bee has experiences at all? We cannot say, but the fascination of the mind-brain question attracts a great many people to research in this area and motivates them to make tireless efforts.



"I think, therefore I am"—René Descartes' declaration points to the mind-body problem facing psychologists: How and why does the brain's activity produce conscious experience?

Psychological Explanations at the Level of the Gene

How do heredity and evolution influence behavior?

The nature-nurture question shows up throughout the field of psychology in one form or another. Why do little boys get into fights more often than little girls do? Is it because of some genetic difference, or is it because other people intentionally or unintentionally encourage boys and girls to act differently? Why do some people develop drug and alcohol-related problems, yet others do not? Why do some people become depressed and others not? We should not expect any of the causes to be entirely genetic or entirely environmental, but it is pointless to say simply “both are important.” Ideally, we would like to know exactly how certain genes influence development, which aspects of environment are most crucial, and how genes and environment combine and interact.

Let’s first take a quick survey of genes and what they do. Then we shall explore in more detail the application of genetics to human behavior. At the end of this module we shall attend to the evolution of behavior.

Principles of Genetics

If you have already studied genetics in a biology class, much of this discussion will be a review. Skim it if you already know this material; read more carefully if it is unfamiliar to you.

Nearly every plant or animal cell (with a few exceptions, such as red blood cells) contains a nucleus, which includes *strands of hereditary material* called **chromosomes** (Figure 3.1). Chromosomes provide the chemical basis of heredity. Humans have 23 pairs of chromosomes in each cell of the body, except egg or sperm cells which have 23 unpaired chromosomes. At fertilization, the 23 chromosomes in the egg cell combine with the 23 in the sperm to form the 23 pairs that will characterize the new person (Figure 3.2).

Sections along each chromosome are known as **genes**, *these control the chemical reactions that direct the individual’s development*, for example, determining whether

one becomes a tall dark-haired woman or a short blond man. Genes exert these effects by controlling the body’s chemistry: Genes are composed of a chemical called DNA, which controls the production of another chemical called RNA, which in turn controls the production of proteins. These proteins either become part of the body’s structure or control the rates of chemical reactions in the body. So, genes exert their effects by a long, complex route (Figure 3.3).

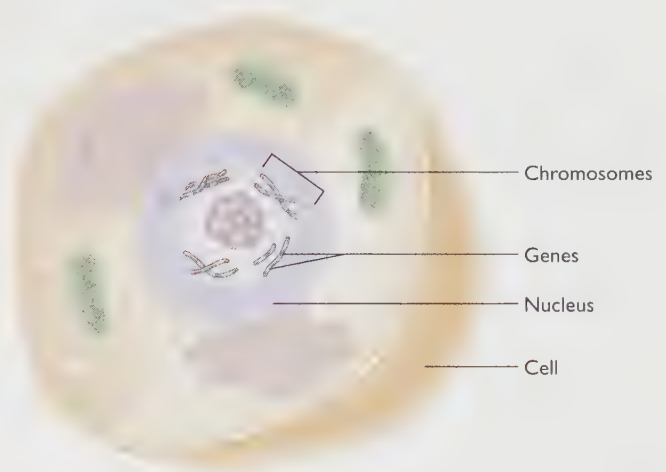


FIGURE 3.1 Genes are sections of chromosomes in the nuclei of cells. (Scale is exaggerated for illustration purposes.)

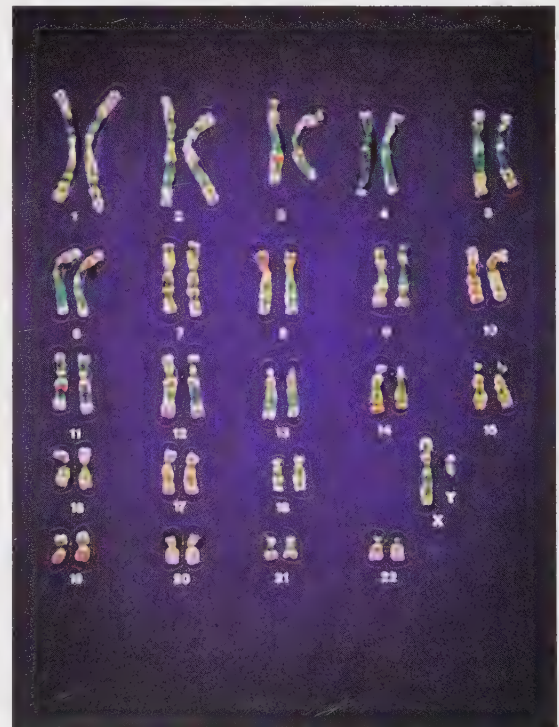


FIGURE 3.2 The nucleus of each human cell contains 46 chromosomes, 23 from the sperm and 23 from the ovum, united in pairs.

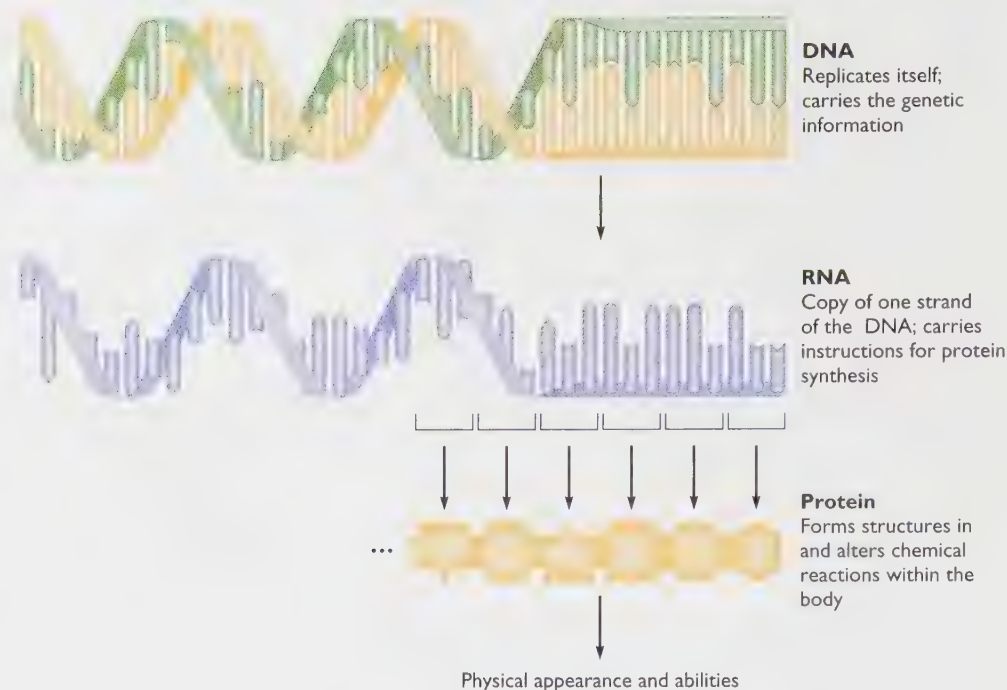


FIGURE 3.3 The genes, composed of DNA, control the production of RNA, which in turn controls the production of proteins. Proteins form many structures of the body (such as muscles); they also control the rate of many chemical reactions (such as digestion).

All body cells that contain pairs of chromosomes have pairs of genes, one on each chromosome. You have two genes for eye color, two for hair color, and so forth. (The exception: Men have one X chromosome and one Y chromosome and therefore have unpaired genes on their X and Y chromosomes.) *If both genes of a pair are the same*, you are **homozygous** (HO-mo-ZI-gus) for that gene. *If the two genes are different*, you are **heterozygous** (HET-er-o-ZI-gus) for that gene (Figure 3.4). (A *zygote* is a fertilized egg.

Homozygous means that the egg—or other cell—has the same gene on both chromosomes.)

If you have one gene for brown eyes and one for blue eyes, you will have brown eyes. The gene for brown eyes is considered a **dominant gene** because *it exerts evident effects even if someone is heterozygous for that gene*; a **recessive gene** shows its effects only in the homozygous condition. Very few behavioral differences in humans depend on a single gene. One example, admittedly not a very

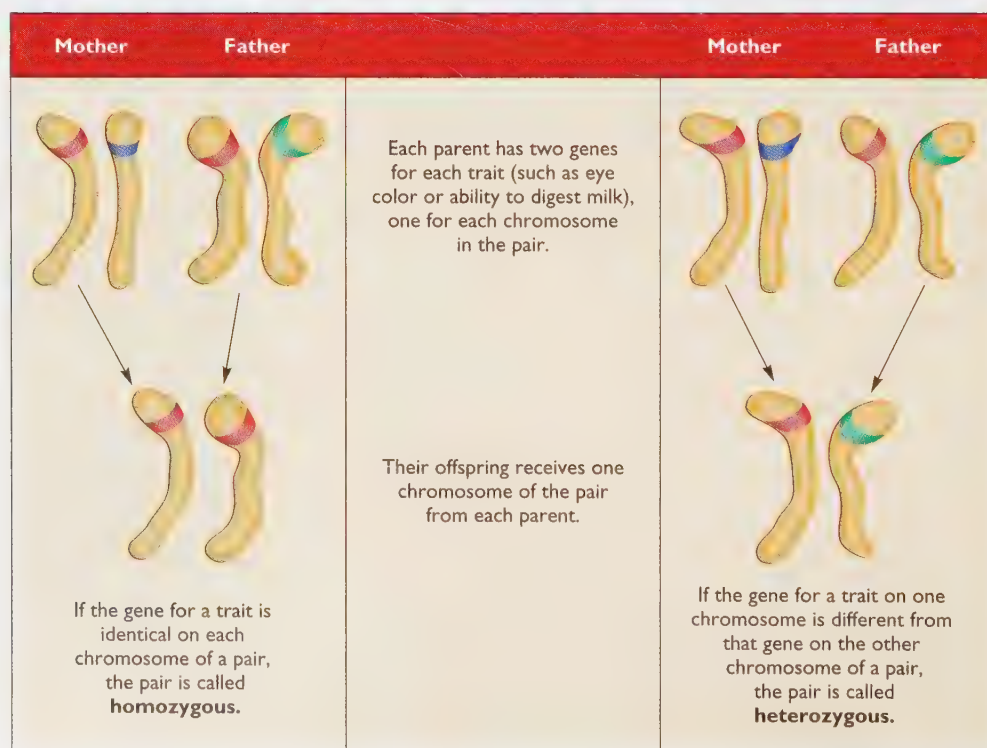


FIGURE 3.4 In a pair of homozygous chromosomes, the gene for a given trait is identical on both chromosomes. In a heterozygous pair, the chromosomes contain different genes for a trait.

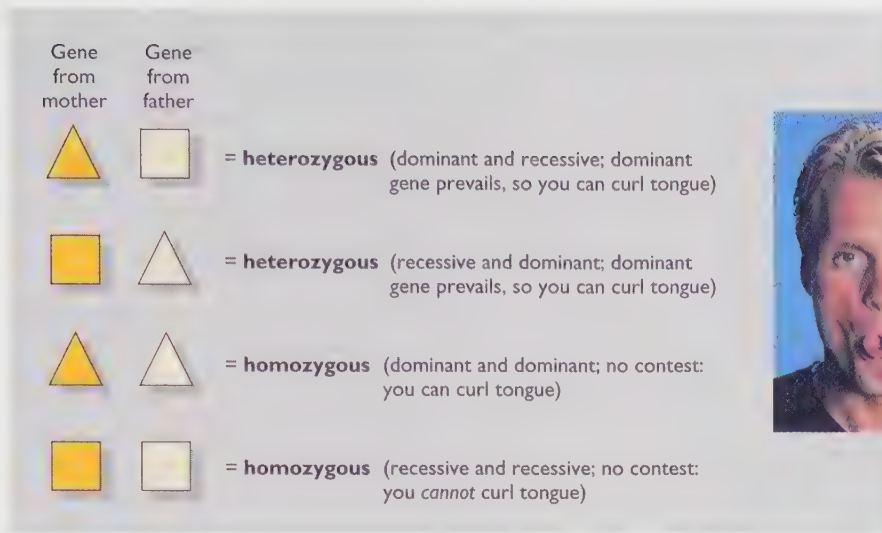


FIGURE 3.5 This figure depicts the ability to curl the tongue lengthwise as an example of a behavior that depends on a single gene. The gene that enables you to curl your tongue is a dominant gene, indicated by a triangle here. The square refers to a recessive gene for the inability to curl the tongue.

important one, is the ability to curl the tongue lengthwise (Figure 3.5). The gene enabling tongue-curling is dominant, so if you have this gene, you can curl your tongue. If you are homozygous for the noncurling gene, you will have to get through life without this ability.

A person who is heterozygous for a particular gene will show the effects of the dominant gene but can still pass on the recessive gene to a son or daughter. For example, two parents who are heterozygous for the tongue-curling gene will both be able to curl their tongues, but they could each pass a recessive gene to their child, who would then lack this ability.

CONCEPT CHECKS

1. Suppose you can curl your tongue but your sister cannot. Are you homozygous or heterozygous for the tongue-curling gene, or is it impossible to say? What about your sister?
2. If two parents cannot curl their tongues, what can you predict about their children? (Check your answers on page 74.)

Sex-Linked and Sex-Limited Genes

One pair of human chromosomes is known as **sex chromosomes**, because they *determine whether an individual will develop as a male or as a female*. There are two types of sex chromosomes, known as X and Y (Figure 3.6). A *female has two X chromosomes* in each cell; a *male has one*



FIGURE 3.6 An electron micrograph of X and Y chromosomes shows the difference in length. (From Ruch, 1984.)

X chromosome and one Y chromosome. The mother contributes one X chromosome to each child, and the father contributes either an X or a Y chromosome.

Genes located on the X chromosome are known as X-linked or sex-linked genes. An X-linked recessive gene shows its effects more often in men than in women. For example, the most common type of colorblindness depends on an X-linked recessive gene and is therefore more common in men than in women. A man with that gene on his X chromosome will definitely be colorblind, because he has no other X chromosome. He has a Y chromosome, but that Y chromosome contains neither the gene for colorblindness nor the gene for normal color vision. A woman who has the colorblindness gene has a second X chromosome, which probably contains a dominant gene for normal color vision. If so, she will have normal color vision herself but could transmit the colorblindness gene to any of her children (Figure 3.7).

Genetically controlled differences between the sexes do not necessarily depend on sex-linked genes. For example, adult men generally have deeper voices and more facial hair than women do. Those characteristics are controlled by genes that are present in both sexes but activated by men's hormones. Similarly, the genes controlling breast development are present in both sexes but activated by women's hormones. **Sex-limited genes** are *those that affect one sex more strongly than the other, even though both sexes have the genes*.

Why do most boys fight more than girls do? We don't know, but if genes are responsible, they are probably *sex-limited* genes rather than sex-linked genes. Male hormones may activate certain genes that promote aggressive behavior; we have no evidence suggesting an X-linked or Y-linked gene for aggressive behavior.

CONCEPT CHECK

3. Suppose a colorblind man marries a woman who is homozygous for normal color vision. What sort of color vision will their children have? (Check your answer on page 74.)

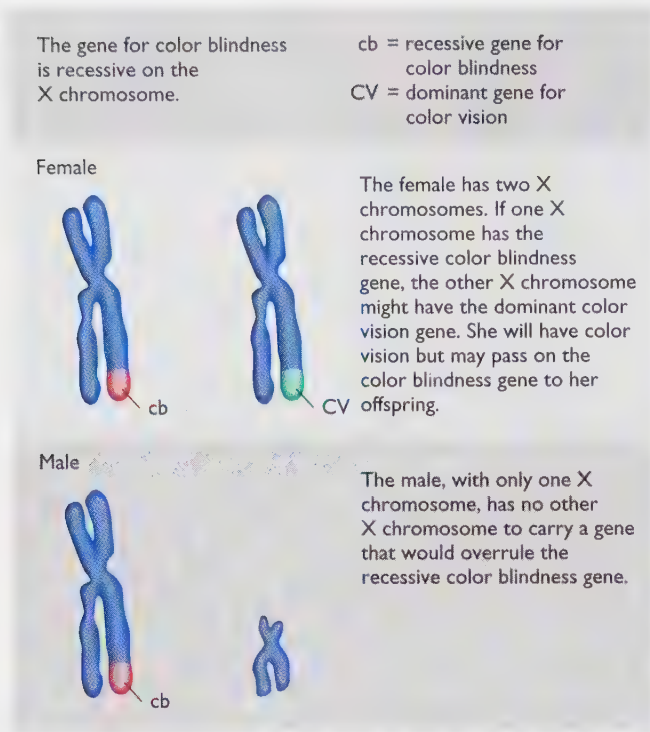


FIGURE 3.7 Why males are more likely than females to be colorblind.

Identifying and Localizing Genes

Modern technology enables researchers to identify and localize the genes responsible for certain conditions. For example, researchers have located several genes that increase the risk of Alzheimer's disease (a condition associated with progressive memory loss), one gene strongly linked with Huntington's disease (associated with loss of muscle control) and genes that lead to several other diseases. You could (for a price) have someone examine your genes today and tell you which diseases you are likely to get, which ones you will not get, and in some cases even an estimate of how soon you will contract a particular disease. As research progresses, perhaps a chromosomal test will someday be able to tell you about your risks for depression, alcoholism, schizophrenia, and other behavioral disorders.

How much would you want to know? For most of us, the answer depends on what we could do with the information. For example, if you learned that you were predisposed to alcoholism, you might take some precautions to prevent it. At a minimum, you might decide not to join a fraternity or sorority with a reputation for heavy drinking. On the other hand, learning that you were likely to develop Alzheimer's disease at age 70 would do you less good, because no one knows yet how to prevent Alzheimer's. Still, identifying the genes that predispose one to Alzheimer's disease may lead researchers to a better understanding of the disease and ultimately to methods of prevention or treatment.

Identifying risk genes poses some possible harms as well. Conceivably, employers or insurance companies could

someday require genetic tests of all job applicants and then discriminate against anyone whose test indicated a likelihood of developing a serious disease or psychological disorder. At present, no employers or insurance companies seem inclined to institute such a policy, but society will certainly face a serious ethical issue.

Measuring Heritability in Humans

The ability to curl your tongue lengthwise is an oddity; it apparently depends entirely on a single gene. Huntington's disease (a condition marked by tremors, loss of muscle control, and eventual deficits in memory and judgment) is a disorder also associated with a single gene: If you have that gene, you will almost certainly get the disease (unless you die of something else first), and if you don't have the gene, you cannot get the disease, as far as we know. We could name a few other examples of single-gene determinants of behavior, but not many.

For the most part, in psychology, we deal with behaviors that develop through complex influences of many genes and many environmental influences. You will occasionally hear someone ask whether a behavior, alcoholism for example, depends on heredity or on environment. That question is meaningless as stated. Obviously, you need both heredity and environment. However, after a small change in wording, the question becomes more meaningful and potentially answerable: Does a given *difference* in behavior (such as that one person becomes an alcoholic and another doesn't) depend more on *differences* in heredity or *differences* in environment? That is, if we want to predict whether more people in group A or group B will become alcoholics, how well could we succeed if we knew their heredity, and how well if we knew their environment?

The answer to a question like this is summarized by the term **heritability**, *an estimate of the variance within a population that is due to heredity*. Heritability ranges from 1, indicating that all variance is due to heredity, to 0, indicating that none of it is. For example, tongue curling has a heritability of almost 1. (It would be exactly 1 except that it is possible to suffer an injury to the tongue muscles that prevents tongue curling.)

For an example of a condition with zero heritability, consider the language that someone speaks. If you want to predict whether a child will speak English, Greek, or Bengali, you should ask which language is spoken by the child's parents and others around the child. An examination of the child's genes will not help at all.

Most behaviors have a heritability somewhere between zero and one. To estimate the heritability of any behavior, researchers rely on the following types of evidence (Segal, 1993):

- Do **monozygotic** (identical; literally, "one-egg") **twins** resemble each other more closely than **dizygotic** (fraternal; literally, "two-egg") **twins** do (Figure 3.8)? Monozygotic twins have identical heredities; dizygotic

twins resemble each other genetically only as much as a brother and sister do. A greater resemblance between monozygotic twins is suggestive but not conclusive evidence for a genetic influence. A *lack* of such resemblance would argue strongly against a genetic influence.

- Do twins who are adopted by separate families and reared apart resemble each other more closely than we would expect for unrelated people? If so, genetic similarity is a likely explanation, although it is not the only possibility. Twins also share a *prenatal* (before-birth) environment; they share influences from the mother's eating, drinking, and smoking habits, her age, and so forth, and those prenatal influences can have a major influence on brain development (Devlin, Daniels, & Roeder, 1997).
- To what extent do adopted children resemble their adoptive parents and their biological parents? If children who are adopted in infancy closely resemble their biological parents in some regard, that resemblance probably reflects a genetic influence.

SOMETHING TO THINK ABOUT

A greater resemblance between a pair of monozygotic twins than between a pair of dizygotic twins is not considered conclusive evidence for a genetic influence. Why not? ★

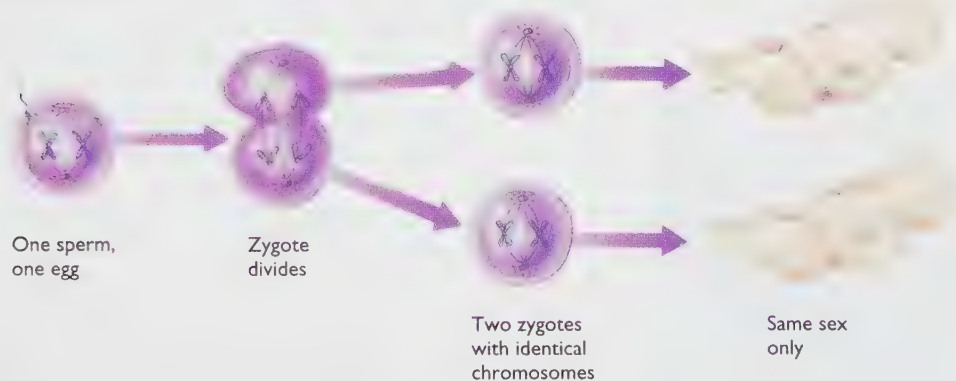
CONCEPT CHECK

4. If our society changed so that an equally good environment was provided for all children, would the heritability of behaviors increase or decrease? (See page 74.)

Whenever possible, researchers try to examine data from both twins and adopted children. Almost any kind of evidence has possible flaws and alternative explanations for its results, but if two or three kinds of studies all indicate a probable genetic influence, the conclusion becomes stronger.

Based on studies of twins and adopted children, researchers have found at least moderate heritability for a wide variety of behaviors and conditions, including schizophrenia, depression, and alcohol abuse (Loehlin, Willerman, & Horn, 1988), as well as a range of personality traits (Nigg & Goldsmith, 1994) and attitudes (Tesser, 1993). Adopted children have been shown to resemble their birth parents more than they do their adopting parents with regard to how much time they spend watching television (Plomin, Corley, DeFries, & Fulker, 1990). Monozygotic twins reared apart resemble each other in their religious devoutness (Waller, Kojetin, Bouchard, Lykken, & Tellegen, 1990). (Their religious affiliation, however, depends on the adoptive family. There is no gene for Presbyterian.)

Identical twins



Fraternal twins

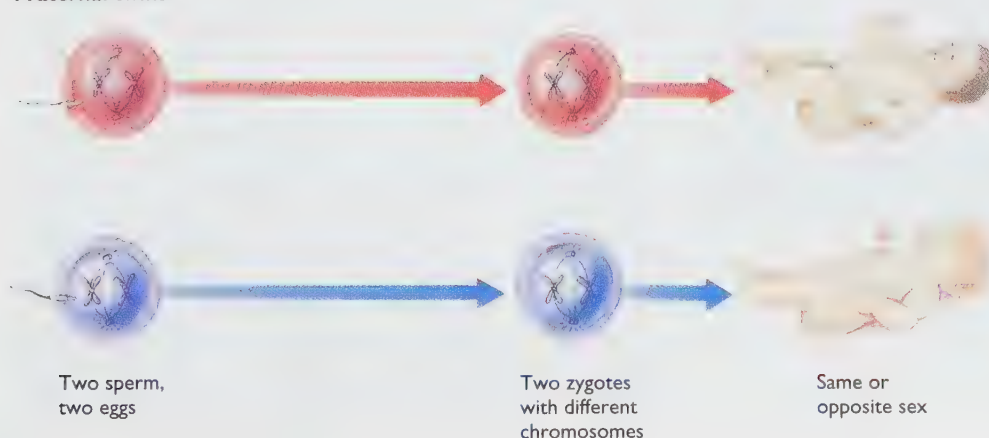


FIGURE 3.8 Identical (monozygotic) twins develop from the same fertilized egg. Fraternal (dizygotic) twins grow from two eggs fertilized by two different sperm.

Monozygotic twins reared apart also resemble each other, sometimes to an amazing degree, in their interests (such as hunting, fishing, or arts and crafts), an optimistic or pessimistic outlook on life, and even their hobbies (Lykken, McGue, Tellegen, & Bouchard, 1992). One pair of twins who were separated at birth, reared in different western Ohio cities, and reunited in adulthood discovered that they had a great deal in common: Both had been named Jim by their adoptive parents. Each liked carpentry and drafting, had built a bench around a tree in his yard, worked as a deputy sheriff, drove a Chevrolet, took his vacations in western Florida, had gained weight at the same age, had married a woman named Linda, had divorced her and married a woman named Betty, and had a son named James Alan (or James Allen) and a pet dog named Toy (Figure 3.9). Surely, some of these similarities are mere coincidences. (Lots of people drive Chevrolets, for example.) And clearly the environment was not irrelevant. If either twin had been adopted by a family in Afghanistan, he would have had a hard time finding a woman named Linda or Betty and he probably would not have taken his vacations in Florida. Nevertheless, so many twins separated from birth have such striking similarities that it is hard to deny that at least part of the similarity reflects genetic influences.

Why might monozygotic twins resemble each other so closely? One possible answer is that many behavioral traits depend on the combined influences of many genes. For example, genes A, B, C, and D might together produce an

effect that we would never suspect from any one of the genes separately, much as the effects of mixing four ingredients might produce a cake that one could never approximate from only three of the ingredients. Because monozygotic twins share all their genes in common, they will share these combinations or configuration results; other people in the same family will not.

The Route from Genes to Behavior

Some years ago, when researchers announced an apparent link between one gene and a predisposition to alcoholism, several psychologists said that this link was impossible, because no mere gene could explain something so complicated as alcoholism. Similar reactions have greeted announcements of genes linked to many other behaviors and personality traits.

Perhaps the problem is that, from the first time we are introduced to genetics, we are given examples such as eye color, in which a single gene has almost complete control of the outcome and environmental variations have almost no effect. With alcoholism and many other behaviors, it is obvious that no one gene can have such tight control. Furthermore, it is more difficult in some cases than others to imagine a route from gene to behavior. Certain genetic variations in humans are known to affect the microscopic structure of certain parts of the brain, in ways that probably affect behavior (Benjamin et al., 1996; Blum, Cull, Braverman, & Comings, 1996; Zuckerman, 1995). But how could genes affect a complicated behavior like television viewing? or religious devoutness? or alcoholism?

Probably no one gene has any direct effects on these behaviors, although genes can influence a behavior by influencing other characteristics that are either associated with it or antagonistic to it. For example, a gene that tends to make someone active and restless will probably interfere with any behavior that requires sitting for long periods—such as watching television or attending religious services.

Or consider dietary choices: Most Asian adults, including Americans of Asian ancestry, seldom drink milk. Within other ethnic groups, some adults enjoy milk, but others do not. Part of the variations in dairy consumption are under genetic control, but the genes do not affect taste or motivation; rather, they control the body's ability to digest *lactose*, the sugar in milk (Figure 3.10).

Almost all infants of any ethnic group can digest lactose. As they grow older, most Asian children and a large number of non-Asian children lose the ability to digest it. (They lose that ability even if they drink milk frequently. The loss depends on genes.) They can still eat and enjoy small amounts of milk, but larger amounts will cause gas and stomach cramps (Flatz, 1987; Rozin & Pelchat, 1988).



FIGURE 3.9 Identical twins Jim Lewis and Jim Springer were separated at birth, reared in separate cities of western Ohio, and reunited in adulthood. Like many identical twins who were reunited after growing up separately, they discovered that they shared a long list of detailed similarities.

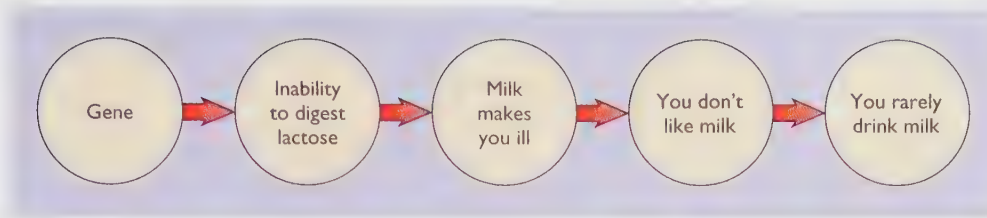


FIGURE 3.10 Genes don't control behavior directly but instead through many indirect routes. They control chemical reactions in the body, which in turn influence behavior.

Figure 3.11 shows how the ability to digest dairy products varies from one part of the world to another. The main point is that one gene can affect a behavior by altering chemical reactions outside the brain itself.

Consider another result: A study of thousands of Australian twins found that monozygotic twins resembled each other more closely than dizygotic twins did with regard to age of first sexual intercourse (Dunne et al., 1997). It is always hazardous to put too much trust in any one study, but for the sake of illustration, let's accept the conclusion that genes influence the age at which one first has sexual intercourse, and then let's ask how a gene could possibly do so. First, obviously one gene does not directly control that behavior, nor does it override the environment. If some of these twins had been adopted into a Bedouin Arab society, their sexual behavior would have been severely constrained. In Bedouin Arab culture, boys and girls are not permitted to play together or to intermingle in any way before marriage. A girl who managed nevertheless to have premarital intercourse would not be permitted to marry—a disaster in a society with hardly any jobs for women. Whatever genes influenced behavior in the Australian study had effects that depended on a culture that tolerates premarital sex.

So, how could a gene influence age at first sexual intercourse? We can imagine several possibilities:

- Genes influence rate of maturation: Those who begin puberty at 12 or 13 are more likely to have early sexual experiences than those who start puberty at 16 or 17.
- Genes influence secretion of sex hormones: Young people with higher hormonal levels probably have a stronger sex drive.
- Genes influence physical appearance: A very attractive individual will probably have more opportunities for sexual activity than someone less appealing.

Note that this third possibility implies that a gene can affect your behavior by influencing how other people react to you. That example underscores the difficulty of separating genetic influences from environmental influences. If you have a gene that affects your appearance and your appearance influences the way other people treat you, then the gene has influenced your behavior. If you had an identical twin who was reared in a separate environment, both of you would no doubt share some similar behaviors, and those similarities would count as evidence for heritability of the behavior; however, demonstrating heritability (even high heritability) for a behavior does not tell us how the genes exert their effects and does not deny that a change in the environment might drastically change the behavior.

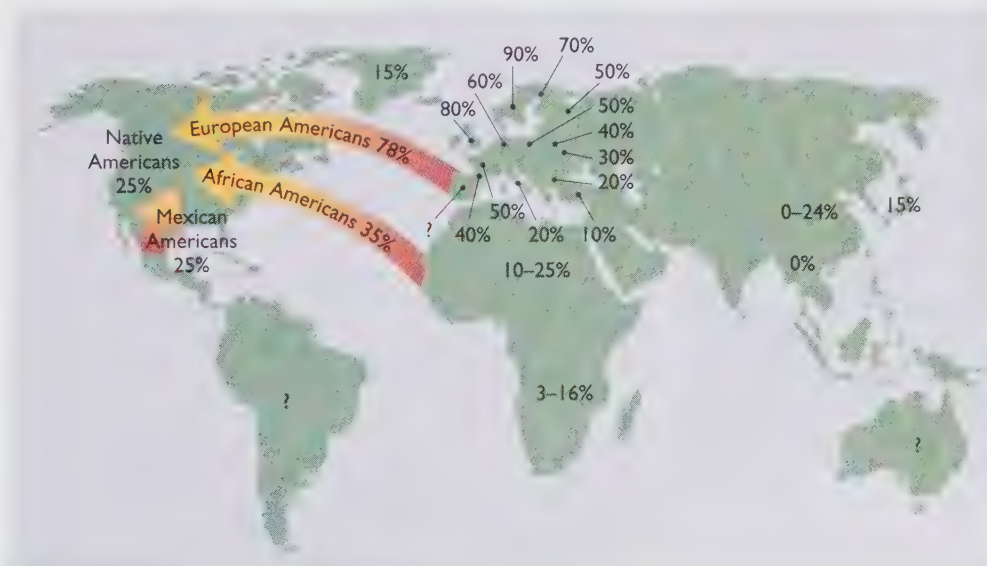


FIGURE 3.11 Adult humans vary in their ability to digest lactose, the main sugar in milk. The numbers in this figure refer to the percentage of each population's adults that can easily digest lactose. In Asian countries and other locations where most adults cannot digest lactose, cooks seldom use dairy products. (Based on Flatz, 1987, and Rozin & Pelchat, 1988.)

How Heredity Interacts with Environment

You might sometimes hear someone say, “I hope researchers never conclude that . . . [fill in the blank: intelligence, depression, alcoholism, etc.] is under genetic control, because if it is, that means that we can’t do anything about it.” That statement is nonsense. The fact that genetic differences have a major influence says nothing about whether some environmental intervention might change the results. For example, even if you have genes for straight hair, your health, your diet, and your grooming habits might cause you to have straight hair, curly hair, or no hair at all.

Here is another example: **Phenylketonuria (PKU)** is an inherited condition that, if untreated, leads to mental retardation. (The condition depends on a recessive gene. About 2% of people of European or Asian ancestry are heterozygous carriers for this gene; the gene is absent in African populations.) People who are homozygous for PKU lack the chemical reactions that break down a substance called *phenylalanine*, a common constituent of proteins, into other chemicals. On an ordinary diet, an affected child will accumulate phenylalanine in the brain and become mentally retarded; however, an affected child who observes a diet low in phenylalanine for at least the first 12 to 15 years of life will not become mentally retarded. Thus, an environmental intervention (here, a controlled diet) can greatly influence a condition that is known to be under genetic control.

Evolution

Our genes are a product of evolution. Given what we know about genes, we can infer that evolution *must* occur and must have been occurring for as long as genetics and reproduction have been as they are today. The argument, based on the ideas of English naturalist Charles Darwin (1859) is as follows:

1. The genes that an organism inherits from its parents strongly influence its characteristics. In short, like begets like. Children generally look somewhat like their parents, for example.
2. On occasion, genetic variations will cause an organism to differ from its parents. Such variations may arise from recombinations of genes (some from one parent and some from the other) or from **mutations**, *random changes in the structure of genes*. Recombinations and mutations alter the appearance or activity of the organism. Most mutations are disadvantageous, although an occasional mutation will give an individual an advantage in coping with some situations.
3. If individuals with a certain gene or gene combination reproduce more successfully than others do, the genes that confer an advantage will spread. Over many generations, the frequency of those genes will increase while the frequency of others will decrease (Weiner, 1994). Such

changes in the gene frequencies of a species constitute evolution. We know that mutations sometimes occur in genes and that an occasional mutation can lead to greater success in reproduction, so we can logically deduce that evolution *must* occur.

This argument does not tell us, however, whether all forms of life evolved from a single common ancestor. To deal with that question, we need other kinds of evidence, such as the fossil record.¹

Animal and plant breeders discovered a long time ago that they could develop new strains through **artificial selection**, or *selective breeding*. By purposefully breeding only animals with certain traits, breeders developed cocker spaniels, thoroughbred racehorses, and chickens that lay enormous numbers of eggs. Darwin’s theory of evolution stated that **natural selection** can accomplish the same results as selective breeding. If, in nature, *individuals with certain genetically controlled characteristics reproduce more successfully than others do, then future generations will come to resemble those individuals more and more.*

Some people assume, mistakenly, that evolution means “the survival of the fittest,” but what really matters for evolution is not survival but *reproduction* (Figure 3.12). Someone who lives to the age of 100 without having a child has

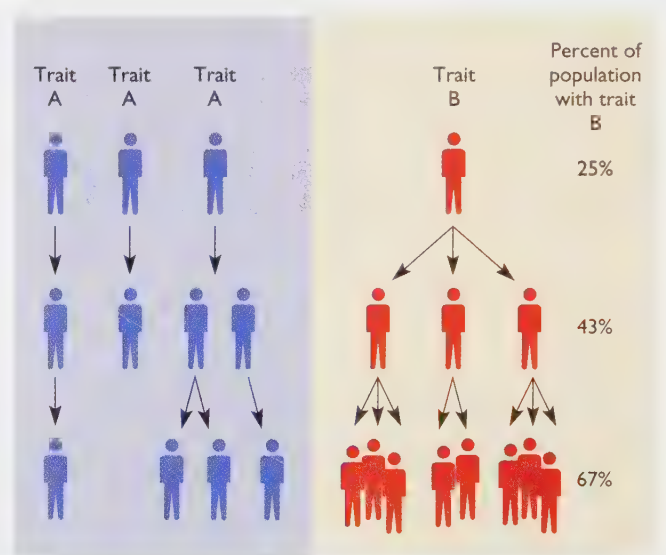


FIGURE 3.12 What’s important for evolution is reproduction, not survival. Here, the population starts with three people carrying trait A and one with trait B. The person with B and his or her descendants produce more children, on the average, than people with A do. Consequently, the genes controlling trait B will increase in prevalence from one generation to the next.

¹What about “creation science?” you may ask. Creation scientists argue that certain observations are hard to explain in terms of natural selection; however, no one has yet found evidence supporting the alternative view, that life began suddenly in more or less its present form. It is fair to admit that biologists do not know everything about how evolution operates, but the points that biologists dispute are the details, not the fundamental concept of natural selection.

failed to spread his or her genes. In contrast, a person who has five healthy children before dying at age 30 is a big success, in evolutionary terms.

A gene that increases a person's chance of surviving long enough to reproduce will be favored over a gene that causes death in infancy. But genes that have no influence on the individual's survival can also be favored. For example, a gene that makes an individual more successful at attracting mates would certainly be favored, as would a gene that makes someone more successful at protecting his or her offspring or other close relatives.

CONCEPT CHECK

5. Infertile worker bees are sisters of the queen bee, which lays all the eggs. In comparison with species in which all individuals are fertile, would you expect worker bees to be more likely or less likely to risk their lives to defend their sister? Would you expect a queen bee to be more or less likely than a worker bee to risk her life? (Check your answers on page 74.)

Occasionally, people say something like “every generation our little toes get smaller and smaller because we don’t use them,” or “through evolution, we will gradually get rid of the human appendix because we don’t need it.” Such statements reflect a misunderstanding of evolution. We do not add, change, or lose genes because of the way we use some part of the body. Our lack of need for an appendix has no effect on the genes controlling the appendix. The only way that people could evolve, say, a reduction in the size of the little toe would be if people with genes for “smaller than average little toe” reproduced more than other people.

One other common misunderstanding: Evolution does not necessarily mean long-term improvement. Genes will spread within a population if they provide some benefit at the moment. The result could be a population that is beautifully adapted to its current environment but ill-adapted when the environment changes in some way.

The Evolution of Behavior

Psychologists learn about the evolution of behavior largely through studies of animals. The study of animal behavior grew out of two separate pursuits: ethology and comparative psychology. **Ethology** is *the branch of biology that studies animal behavior under natural or nearly natural conditions*. Ethologists emphasize **species-typical behaviors**—*behaviors that are widespread in one animal species but not in others*. Species-specific behaviors are sometimes described as *instinctive*, although many investigators shun that term. (Many people use the term *instinct* as though it constituted an explanation. For example, they might say that a mother squirrel takes care of her young because of her “maternal instinct.” But that statement is like saying “she

does it because she does it.” Calling something an instinct tells us nothing about how the behavior develops or what processes control it.)

Comparative psychology is *the branch of psychology that compares the behaviors of various animal species*. A comparative psychologist might study which species are best at localizing sounds, how species differ in their means of finding food, or why some species solve a particular problem faster than other species do. Both ethologists and comparative psychologists study how behavioral capacities evolve and how they develop within an individual.

Evolution adapts the behavior of an animal, as well as its anatomy, to its way of life. Consider the mating behavior of the kittiwake, a member of the gull family (Tinbergen, 1958). Kittiwakes, unlike other gulls, nest on narrow ledges of steep cliffs (Figure 3.13). Because there are only so many suitable ledges, kittiwakes fight ferociously to claim territories. By contrast, herring gulls, which nest on the ground, rarely fight over territory, because for them one nesting site is about as good as any other. Kittiwakes use mud to build a hard nest with a barrier to prevent their eggs from rolling off the ledge. Herring gulls make no such effort. When kittiwake chicks hatch, they remain virtually motionless until they are old enough to fly. The advantage of this behavioral tendency is clear: A chick that takes even a step or two may fall off the ledge. Herring gull chicks, in contrast, begin to wander out of their nest long before they can fly.

Each of these kittiwake behaviors—fighting over territory, building secure nests, and remaining motionless—is well adapted to life on a narrow ledge. But have these behaviors been built into the animal by evolution, or are they learned anew by each individual? In the rare cases when kittiwakes nest on the ground, the chicks remain motionless



FIGURE 3.13 The nesting behavior of kittiwakes is superbly adapted for their survival. For example, the parents build a mud barrier on the edge of the nest, and the young remain motionless until they are able to fly.

anyway, even though they are in no danger of falling. If the egg of a herring gull is placed in a kittiwake's nest, the kittiwakes accept the foreign egg and care for the chick after it hatches. But the chick invariably takes a few steps and falls to its death. Evidently, some behavioral differences are products of the evolution of each species, rather than learned behavior.

Sociobiology

Sociobiology is a field that tries to relate the social behaviors of a species to its biology, particularly to its evolutionary history. According to sociobiologists, an animal interacts with others of its species in a particular way because similar behaviors in past generations have increased the probability of survival and mating. That is, individuals with a genetic tendency to engage in these social behaviors passed on their genes; individuals that behaved another way were less successful at passing on their genes.

Animal Examples of Sociobiological Explanations

Sociobiologists try to understand how various social behaviors may have helped animals to survive and reproduce. Here are two examples:

- Lions generally live in prides made up of one adult male, several adult females, and their young. If a new male succeeds in driving off the old male, he is likely to kill all the young. Why? Female lions are not sexually receptive while they are nursing their young. By killing the young, the new male brings the females into sexual receptivity and increases the likelihood of spreading his genes (Wilson, 1975).
- In a species of bird called reed buntings, a male-female pair will stick together fairly closely, although some females will also occasionally mate with neighboring males. Males help with incubating the eggs and taking care of the young; however, some males help more than others do. Researchers have found that females that engage in a large number of "extramarital affairs" with neighboring males elicit the least help from their male partners (Dixon, Ross, O'Malley, & Burke, 1994). That is, apparently a male will work hard to help raise the young if the probability is high that he is their father. (We do not assume that he understands all this; evolution has simply prepared him to be more helpful to a faithful mate. Males who behaved this way in the past were more likely to pass on their genes.)

Speculations About Human Sociobiology

Human social behavior is also partly the product of our evolutionary history. In principle, that result is a logical necessity, yet citing precise examples is difficult because we are less

certain about which human behaviors are strongly influenced by our genes and which ones are learned from our culture. Consider a couple of speculative, controversial examples:

- People will work very hard to help one another, sometimes even risking their own lives to help other people. *Doing something to help others with no direct benefit to oneself* is called **altruistic behavior**. Similarly, other species engage in some behaviors that appear to be altruistic; for example, a goose that sees a hawk overhead utters an alarm call that warns other geese. A sociobiological explanation for this behavior is that selection has favored genes that promote altruistic behavior toward one's relatives (Trivers, 1972). That is, individuals that help their offspring and other relatives tend to spread their genes, simply because those relatives have many of the same genes as the altruistic individual. And a mechanism that causes one to help relatives may accidentally cause one to help nonrelatives, too.
- Men are more likely than women are to seek multiple sexual partners. A sociobiological interpretation is that a man who impregnates several women is spreading his genes. A woman gains no such advantage in taking multiple sexual partners. Moreover, if she were to do so, she might have trouble getting any one of them to help her rear the children. Therefore, the sociobiologists say, we may have evolved some sex-limited genes that increase males' interest or decrease females' interest in multiple partners (Buss, 1994).

Human sociobiology is controversial, largely because its adherents sometimes seem to be saying, "This is the way people behave; therefore, they must have evolved a tendency to behave this way, and because we have evolved this way, it is natural and almost necessary that we continue to behave this way." That political message does not logically follow from the science. Even if we understand the biological or historical factors that influence us to act in a certain way, we will not necessarily want to continue acting in that way.

Furthermore, sociobiological theories are, in many cases, difficult to test. Sociobiologists know little about the genes that affect human social behaviors, and they can make only uncertain inferences about life in early human societies. Their best approach is to compare cultures. If they find, say, that a male-female difference is consistent across cultures and from one historical era to another, then the difference is probably a real part of human nature. Sociobiological explanations are as appropriate here as they are for lions or reed buntings. However, if a behavior differs greatly across cultures, then we should be skeptical of a sociobiological explanation. Unfortunately for theory-testers, cross-cultural comparisons often show a combination of consistencies and contrasts among cultures. For example, men in every known culture are more interested than women are in multiple sex partners (a consistency across cultures), but the difference between men and women varies from large in some cultures to slight in others (a difference across cultures). In short, human social behavior depends on a combination of evolutionary and cultural

influences, and psychologists must try to give both influences due recognition. We shall return to this issue in the section on attraction in Chapter 14.

THE MESSAGE

Genes, Evolution, and Behavior

Many people justifiably criticize the concept of *genetic determinism*, the idea that people may be fated to become criminals, alcoholics, or mental patients because of their genes. Certain claims in sociobiology attract similar criticisms. As you have seen in this module, only a few behavioral differences are rigidly determined by genetics. Genes do fully determine whether you can curl your tongue lengthwise. They also largely determine whether you will get Huntington's disease, Alzheimer's disease, and several other diseases, although future research may find ways to prevent these conditions.

For most behaviors of great interest to psychologists, however, genes alter some aspect of the body (not necessarily the nervous system) so that one behavior or another becomes more likely. It is simply wrong to say that genetic or evolutionary influences on behavior imply that behavior is unmodifiable or that the environment is irrelevant. In fact, understanding how the genes exert their effects may give researchers their best clue about which environmental interventions would be most effective for modifying the behavior.

S U M M A R Y

- ★ **Genes.** Genes, which are segments of chromosomes, control heredity. Because chromosomes come in pairs, every person has two of each gene, one received from the father and one from the mother. The sex chromosomes are an exception; each male has one Y chromosome and one unpaired X chromosome. (page 63)
- ★ **Dominant and recessive genes.** A dominant gene exerts its effects even in people who have only one dominant gene, but people must have two of a recessive gene, one on each chromosome, in order to show its effects. (page 64)
- ★ **Sex-linked and sex-limited genes.** Genes on the X chromosome are sex linked. A sex-linked recessive gene will show its effects more frequently in males than in females. A sex-limited gene is present in both sexes, but it exerts its effects more strongly in one than in the other. (page 65)
- ★ **Identifying and localizing genes.** Researchers have located the gene responsible for Huntington's disease, and they are trying to identify and localize as many other human genes as possible. Identifying people's behavioral tendencies and risks by examining their genes offers the potential for significant benefits and also for possible misuse. (page 66)
- ★ **Evidence for genetic influences.** Researchers determine the contribution of genes to human behavior by

studying whether monozygotic twins resemble each other more than dizygotic twins do, by comparing monozygotic twins reared in separate environments, and by examining how adopted children resemble their biological parents and their adoptive parents. (page 66)

★ **How genes affect behavior.** Genes apparently affect an amazing variety of behaviors; however, each gene ordinarily affects a great variety of behaviors, not just one, and a gene can affect behavior indirectly, not necessarily by influencing brain development. (page 68)

★ **Influence of the environment on gene expression.** It is possible for a behavior that reflects a genetic influence to be highly modified by a change in the environment. (page 70)

★ **Evolution.** Evolution by natural selection is a logical necessity, given the principles of heredity and the fact that individuals with certain genes leave more offspring than do individuals with other genes. (page 70)

★ **Study of animal behavior.** Ethologists and comparative psychologists study animal behavior and try to understand how it evolved. (page 71)

★ **Sociobiology.** Sociobiologists try to explain social behaviors in terms of the survival and reproductive advantages of those behaviors. To interpret human behavior in such terms, one must carefully distinguish behaviors that are part of human nature from those that are highly dependent on culture and individual experience. In many cases, we simply do not know enough yet to make that distinction. (page 72)

Suggestions for Further Reading

- Tinbergen, N. (1958). *Curious naturalists*. New York: Basic Books. One of the best books for stimulating interest in animal behavior.
- Weiner, J. (1994). *The beak of the finch*. New York: Knopf. Not really about behavior but an excellent book for its explanations of how evolution works.

Terms

- chromosome** a strand of hereditary material found in the nucleus of a cell (page 63)
- gene** a segment of a chromosome that controls chemical reactions that ultimately direct the development of the organism (page 63)
- homozygous** having the same gene on both members of a pair of chromosomes (page 64)
- heterozygous** having different genes on a pair of chromosomes (page 64)
- dominant gene** a gene that will exert evident effects on development even in a person who is heterozygous for that gene (page 64)
- recessive gene** a gene that will affect development only in a person who is homozygous for that gene (page 64)
- sex chromosomes** the pair of chromosomes that determine whether an individual will develop as a female or as a male (page 65)

- X chromosome** a sex chromosome; females have two per cell and males have only one (page 65)
- Y chromosome** a sex chromosome; males have one per cell and females have none (page 65)
- sex-linked gene** a gene located on the X chromosome (page 65)
- sex-limited gene** a gene that affects one sex more strongly than the other, even though both sexes have the gene (page 65)
- heritability** an estimate of the variance within a population that is due to heredity (page 66)
- monozygotic twins** (literally, “one-egg” twins) identical twins who develop from the same fertilized egg (page 66)
- dizygotic twins** (literally, “two-egg” twins) fraternal twins who develop from two eggs fertilized by two different sperm. Dizygotic twins are no more closely related than are any other children born to the same parents. (page 66)
- phenylketonuria (PKU)** an inherited disorder in which a person lacks the chemical reactions that convert a nutrient called phenylalanine into other chemicals; unless the diet is carefully controlled, the affected person will become mentally retarded (page 70)
- mutation** a random change in the structure of a gene (page 70)
- evolution** changes in the gene frequencies of a species (page 70)
- artificial selection** the purposeful breeding, by humans, of animals with certain traits, also known as selective breeding (page 70)
- natural selection** the tendency, in nature, of individuals with certain genetically controlled characteristics to reproduce more successfully than others do; future generations will come to resemble those individuals more and more (page 70)
- ethology** the branch of biology that studies animal behavior under natural or nearly natural conditions (page 71)
- species-typical behavior** a particular behavior that is widespread in one animal species but not in others (page 71)
- comparative psychology** the branch of psychology that compares the behaviors of various animal species (page 71)
- sociobiology** a field that tries to relate the social behaviors of a species to its biology, particularly to its evolutionary history (page 72)
- altruistic behavior** behavior that benefits others without directly benefiting the individual exhibiting that behavior (page 72)

Answers to Concept Checks

1. It is impossible to say whether you are homozygous or heterozygous for the tongue-curling gene. Because it is a dominant gene, it produces the same effects in both the homozygous and heterozygous conditions. Your sister, however, must be homozygous for the noncurling gene. (page 65)
2. Because both of the parents must be homozygous for the inability to curl their tongues, they can transmit only “non-curler” genes, and their children will be noncurlers also. (page 65)
3. The woman will pass a dominant gene for normal color vision to all her children, so they will all have normal color vision. The man will pass a gene for deficient color vision on his X chromosome; the daughters will therefore be carriers for color blindness. (page 65)
4. If all children had equally supportive environments, the heritability of behaviors would *increase*. Remember, heritability refers to how much of the difference among people is due to hereditary variation. If the environment is practically the same for all, then environmental variation cannot account for much of the variation in behavior. Whatever behavioral variation still occurs would be due mostly to hereditary variation. (Note one implication: Any estimate of heritability applies only to a given population. In another population, the heritability could be much different.) (page 67)
5. Because the infertile worker bees cannot reproduce, the only way they can pass on their genes is by helping the queen bee. Consequently, they will sacrifice their own lives to defend the queen. They will also risk their lives to defend other workers in the hive, because these workers also try to defend the queen. The queen, however, will do little to defend a worker, because doing so would not increase her probability of reproducing. (page 71)

Web Resources

Human Genome Project

www.ornl.gov/TechResources/Human_Genome/home.html

This is the definitive site for understanding the Human Genome Project, from the basic science to ethical, legal, and social considerations, to the latest discoveries.

Explanations of Behavior at the Level of the Neuron

Can we explain our experiences and our behavior in terms of the actions of single cells in the nervous system?

Your brain, which controls everything you do, is composed of cells. Does this mean that every one of your experiences—every sight, every sound, every thought—represents the activity of cells in your brain?

One highly productive strategy in science is *reductionism*—the attempt to explain complex phenomena by reducing them to combinations of simpler components. Biologists explain breathing, blood circulation, and metabolism in terms of chemical reactions and physical forces. Chemists explain chemical reactions in terms of the properties of the 92 naturally occurring elements. Physicists explain the structure of the atom and the interactions among atoms in terms of a few fundamental forces.

Does reductionism apply to psychology? Can we explain human behavior and experience in terms of chemical and electrical events in the brain? Here we attempt to answer these questions.

The Cells of the Nervous System

You experience your “self” as a single entity that senses, thinks, and remembers. And yet neuroscientists have found that the nervous system responsible for your experiences consists of an enormous number of separate cells. The brain processes information in **neurons** (NOO-rons), or *nerve cells*. Many of the neurons in the human nervous system are extremely small; the best current estimate is that the nervous system contains nearly 100 billion neurons (Williams & Herrup, 1988), as shown in Figure 3.14. The nervous system also contains another kind of cells called **glia** (GLEE-uh), *which support the neurons in many ways such as by insulating them and by removing waste products*. The glia are about one tenth of the size of neurons but about 10 times more numerous. Until the early 1900s,

many researchers thought it likely that all the neurons physically merged, that the tip of each neuron actually joined the next neuron. We now know that it does not; each neuron remains separate. How do so many separate neurons and glia combine forces to produce the single stream of experiences that is you?

The secret is communication. Each neuron receives information and transmits it to other cells by conducting electrochemical impulses. Sensory neurons carry information from the sense organs to the central nervous system; neurons of the central nervous system process that sensory information and compare it to past information; motor neurons convey commands to the muscles and glands. Within the central nervous system, each neuron sends information to many others, and these in turn send information to still others; eventually, some of those neurons may send information back to the first one. Out of all this rapid exchange of information, about 100 billion neurons produce a single functioning system.

To understand our nervous system, we must first understand the properties of both the individual neurons and the connections among them. Neurons have a variety of shapes, depending on whether they receive information from a few sources or from many and whether they send impulses over a short distance or over a long distance (Figure 3.15).

A neuron consists of three parts—a cell body, dendrites, and an axon (Figure 3.16). The **cell body** contains the **nucleus** of the cell. The **dendrites** (from a Greek word meaning “tree”) are *widely branching structures that receive transmissions from other neurons*. The **axon** is a *single, long, thin, straight fiber with branches near its tip*. Some vertebrate axons are covered with *myelin*, an insulating sheath that speeds up the transmission of impulses

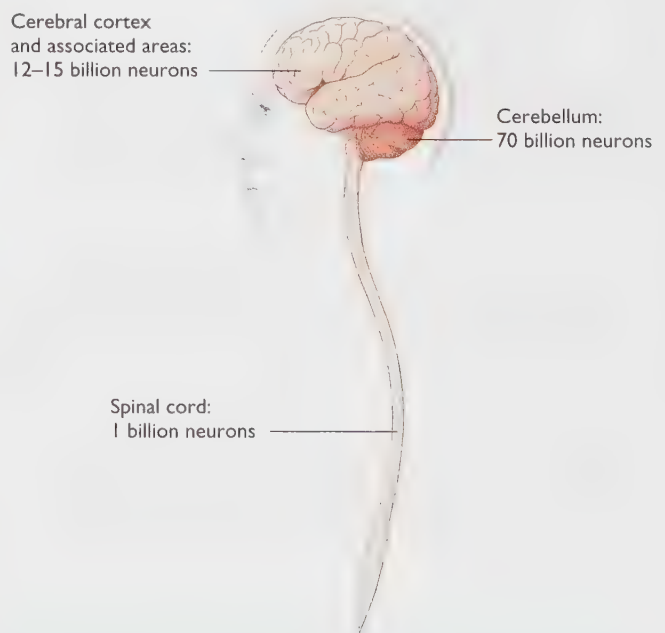
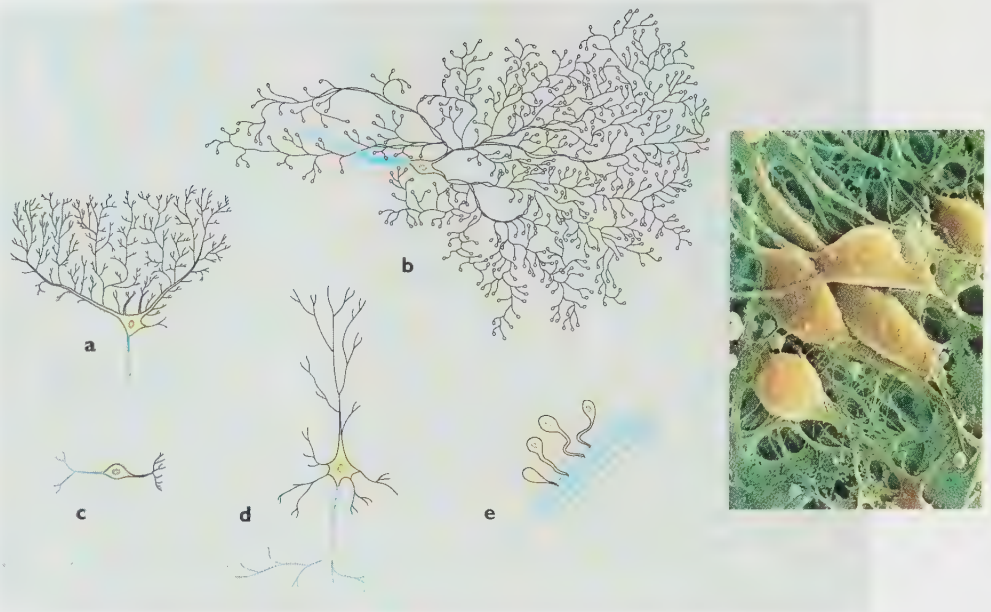


FIGURE 3.14 Distribution of the estimated 83–86 billion neurons in the adult human central nervous system (Based on data of Williams & Herrup, 1988).

FIGURE 3.15 Neurons vary enormously in shape. In each case, the neuron consists of its cell body and all the branched attachments, called axons and dendrites. The neurons in (a) and (b) receive input from many sources, the neuron in (c) from only a few sources, and the neuron in (d) from an intermediate number of sources. The neurons in (e) are sensory neurons, which carry messages from sensory receptors to the brain or spinal cord. The axons are coded blue for easy identification. [Part (b) courtesy of Richard Coss.]



along an axon. As a rule, an axon transmits information to other cells, and the dendrites or cell body of each cell receives that information. That information can be either excitatory or inhibitory; that is, it can increase or decrease the

probability that the next cell will send a message of its own. Inhibitory messages are important for many purposes. For example, during a period of painful stimulation, your brain has mechanisms to inhibit further sensation of pain.

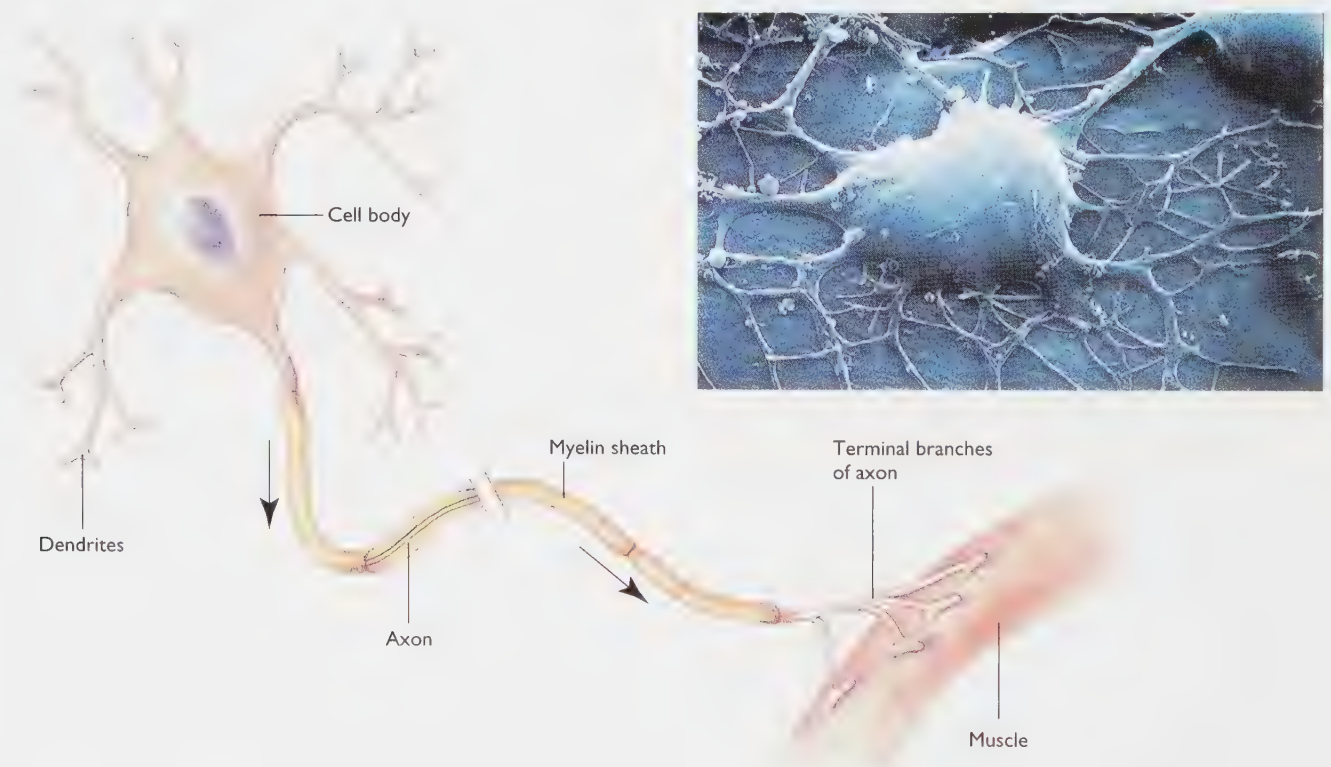


FIGURE 3.16 The generalized structure of a motor neuron shows the dendrites, the branching structures that receive transmissions from other neurons, and the axon, a single, long, thin, straight fiber with branches near its tip. Axons range in length from 1 millimeter to more than 1 meter and carry information to other cells. (inset) A photomicrograph of a neuron.

The Action Potential

Imagine what would happen if axons conveyed information by electrical conduction: Because axons are made of biological materials and not copper wire, the electrical impulse would greatly decrease in intensity as it traveled along the axon. A pinch on your nose would reach the brain at nearly its full intensity; a pinch on your stomach would feel much weaker and a pinch on your toe would feel weaker yet. Short people would be able to feel a pinch on the toe more intensely than tall people would.

So, axons, instead of transmitting electrical impulses, convey information by a special combination of electrical and chemical processes called an **action potential**, *an excitation that travels along an axon at a constant strength, no matter how far it must travel*. An action potential is a yes/no or on/off message. That is, an axon either produces an action potential or it doesn't; it cannot produce larger or smaller or faster or slower action potentials at different times. This principle is known as the *all-or-none law*.

The advantage of an action potential over simple electrical conduction is that action potentials from distant places like your toes reach the brain at full strength, instead of getting weaker as they travel. The disadvantage is that impulses reach your brain more slowly than they would by electrical conduction. Your brain's knowledge is constantly almost one tenth of a second behind what is happening to your toes. Fortunately, that delay does not inconvenience you very often.

To explain how the action potential works, let's start with the resting condition of the axon: Ordinarily, there is *an electrical polarization across the membrane (or covering) of an axon, with a negative charge on the inside of the axon*. This electrical polarization, called the **resting potential** of the membrane, amounts to about 70 millivolts in a typical cell. It is maintained by the different distributions of sodium and potassium ions across the membrane. A mechanism called the sodium-potassium pump keeps sodium more concentrated outside the cell and potassium more concentrated inside. As you probably learned in high-school chemistry, both sodium ions and potassium ions have a +1 charge, so you might imagine that the sodium and potassium effects would cancel each other out; however, the body has a great deal more sodium than it has potassium, so the distribution of sodium makes more difference than the distribution of potassium. The net result is that the outside of the axon (where most sodium is located) is relatively positive; the inside is relatively negative (see Figure 3.17). (Naturally, sodium and potassium are not the only ions present; there are also positively charged calcium ions, negatively charged chloride, and other types as well.)

Certain kinds of stimulation can disrupt this stable resting potential. The action potential occurs when the membrane briefly opens little gates (called sodium gates) along its surface to allow sodium ions to enter the axon (Figure 3.18). As those sodium ions cross the membrane, they

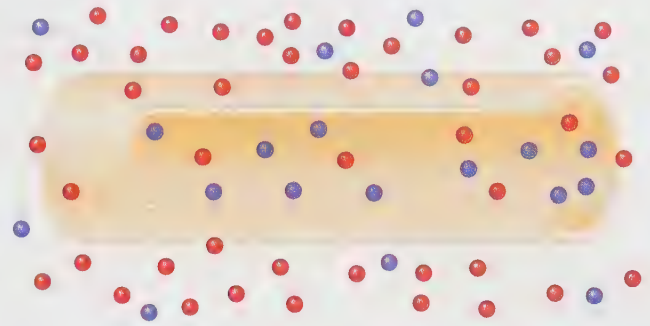


FIGURE 3.17 The axon membrane has a resting potential, when it is not conveying an action potential: the outside is relatively positive and the inside is relatively negative. This difference in electrical potential depends on the distribution of positive ions. The more numerous sodium ions (Na^+) are located mostly outside the axon, whereas the less numerous potassium ions (K^+) are located mostly inside. Here, red spheres represent sodium; blue spheres represent potassium.

eliminate the negative potential that is usually present inside the cell. Then the sodium gates close and potassium gates open to allow potassium ions to leave the axon, returning the axon to its original resting potential (Figure 3.19b). Eventually, the cell pumps the invading sodium ions back out and the lost potassium ions back in.

You will recall that the axon does not conduct like an electrical wire. The action potential travels down the axon like a wave of energy, and the stimulation at each point excites the next point along the axon to create its own action potential. You could imagine the process something like a fire burning along a string: The fire at each point along the string ignites the next point, which in turn ignites the next point. That is, after sodium ions cross at one point along an axon, the electrical

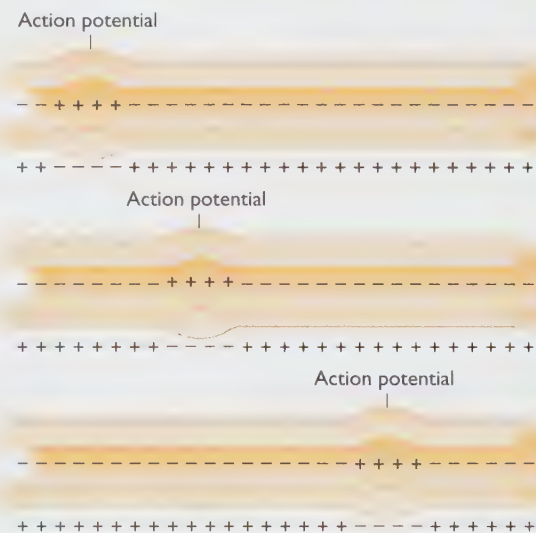


FIGURE 3.18 Ion movements conduct an action potential along an axon. At each point along the membrane, sodium ions enter the axon and alter the distribution of positive and negative charges. As each point along the membrane returns to its original state, the action potential flows to the next point.

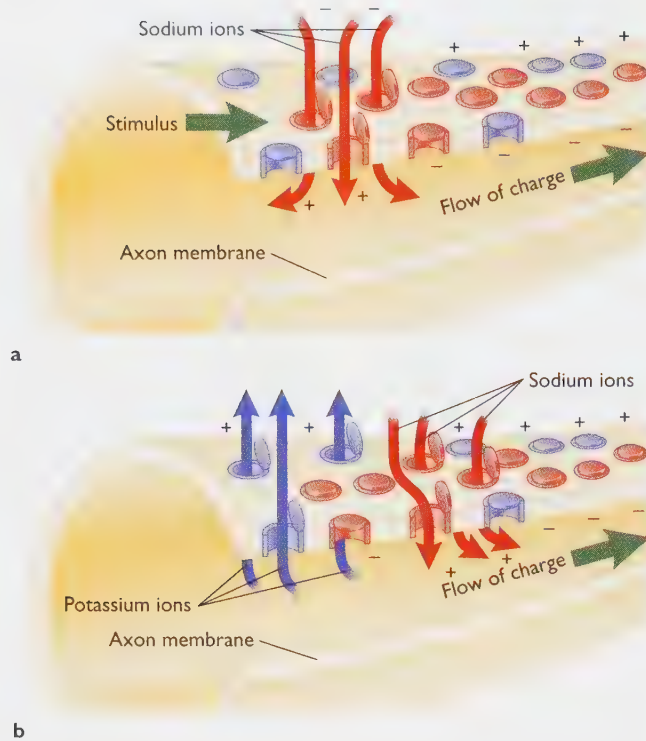


FIGURE 3.19 (a) During an action potential, sodium gates in the neuron membrane open, and sodium ions enter the axon, bringing a positive charge with them. (b) After an action potential occurs at one point along the axon, the sodium gates close at that point and open at the next point along the axon. When the sodium gates close, potassium gates open, and potassium ions flow out of the axon, carrying a positive charge with them. (Modified from Starr & Taggart, 1992.)

excitation opens channels at the next point along the axon, thus enabling sodium ions to cross there, and so forth, as shown in Figure 3.19. In this manner, the action potential remains equally strong all the way to the end of the axon.

Now, all of this information is clearly important to investigators of the nervous system, but why is it important to you as a general psychology student? First, it explains why sensations on points on your fingers and toes do not fade away by the time they reach your brain. Second, an understanding of action potentials is one step toward understanding the communication between one neuron and the next. Third, certain drugs operate by blocking action potentials. For example, anesthetic drugs (such as Novocaine) silence neurons by clogging the sodium gates (van Dyke & Byck, 1982). When your dentist drills a tooth, the receptors in your tooth send out the message “Pain! Pain! Pain!” But that message does not get through to the brain, because a shot of Novocaine has blocked the sodium gates and thereby halted the sensory messages.

CONCEPT CHECK

6. If you stub your toe, do you feel it immediately or is there a delay before you feel it? Why? (Check your answer on page 84.)

Synapses: The Junctions Between Neurons

Ultimately, each neuron must communicate with other neurons. Communication between one neuron and the next follows a different process from the transmission along an axon. At a **synapse** (SIN-aps), *the specialized junction between one neuron and another* (Figure 3.20), one neuron releases a chemical that either excites or inhibits the next neuron. That is, the chemical can make the next neuron either more or less likely to produce an action potential. The events at synapses are central to everything that your brain does, because the synapses are the source of information for the receiving cell.

A typical axon has several branches; each branch ends with a little bulge called a *presynaptic ending*, or **terminal bouton** (or **button**), as shown in Figure 3.21. When an action potential reaches the terminal bouton, it causes the release of molecules of a **neurotransmitter**, *a chemical that has been stored in the neuron and that can activate receptors of other neurons* (Figure 3.21). Different neurons use different chemicals as their neurotransmitters, but each neuron uses the same chemical or same combination of chemicals at all times and at all branches of its axon (Eccles, 1986). The neurotransmitter molecules then diffuse across a narrow gap to *the neuron on the receiving end of the synapse*, the **postsynaptic neuron**. There, the neurotransmitter molecules attach to receptors, which can be located either on the neuron’s dendrites or cell body or (for special purposes) on the tip of its axon. The neural communication process is summarized in Figure 3.22.

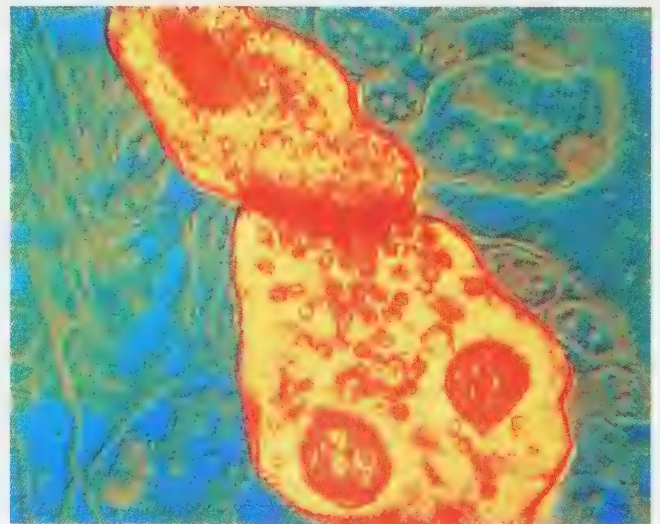


FIGURE 3.20 A synapse, magnified thousands of times in an electron micrograph, includes small round structures in the middle cell called synaptic vesicles, which store neurotransmitter molecules. The thick, dark area at the bottom of the cell is the synapse.

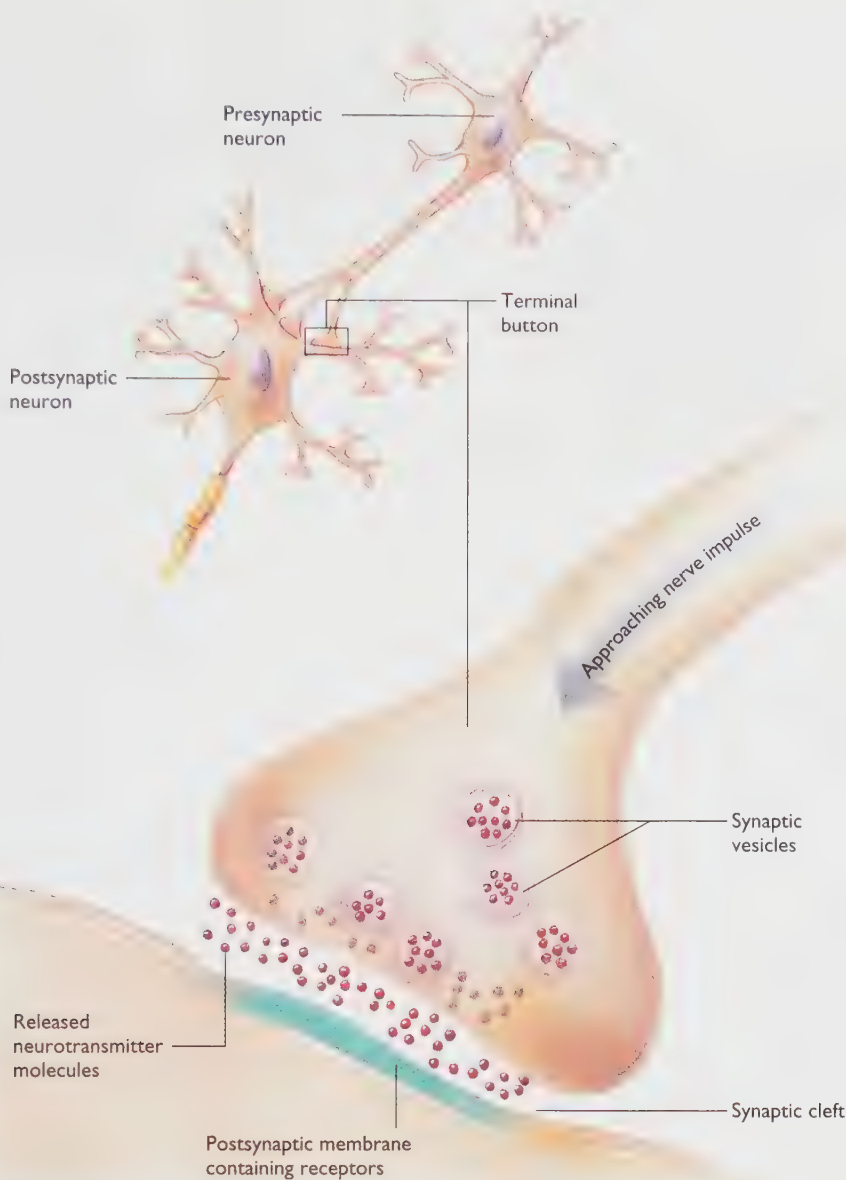


FIGURE 3.21 The synapse is the junction of the presynaptic (message-sending) cell and the postsynaptic (message-receiving) cell. At the end of the presynaptic axon is the terminal bouton (or button), which contains many molecules of the neurotransmitter, ready for release.

Depending on the chemical used as a neurotransmitter and on the type of receptor, the result can be either excitation or inhibition of the neuron. The postsynaptic neuron can receive nearly simultaneous excitation and inhibition from a great many other neurons. This neuron produces an action potential of its own if the total amount of excitation outweighs the total amount of inhibition. That is, positively charged ions can be flowing into the cell at some locations and out at others, while negatively charged ions are also flowing in one direction or the other. The net effect of all these flows can either increase or decrease the frequency of action potentials. The process resembles making a decision: When you are trying to decide whether to do something, you weigh all the pros and cons and then you act if the pros outweigh the cons.

Note that inhibition is not merely the absence of exci-

tation; it is an active braking process. For example, when a pinch on your foot causes you to raise it, thus contracting one set of muscles, inhibitory synapses in your spinal cord block activity in the opposing set of muscles, those that would move your leg the opposite direction. Those inhibitory synapses prevent your spinal cord from sending messages to raise your leg and extend it at the same time.

After a neurotransmitter excites or inhibits a receptor, it displaces from the receptor, terminating the message. From that point on, the fate of the receptor molecule varies. It could become reabsorbed by the axon that released it (through a process called *reuptake*); it could diffuse away, get metabolized, and eventually show up in the blood or urine; or it could bounce around for a moment, return to the postsynaptic receptor and reexcite it. Many drugs, ranging from antidepressants to drugs of abuse, act by blocking

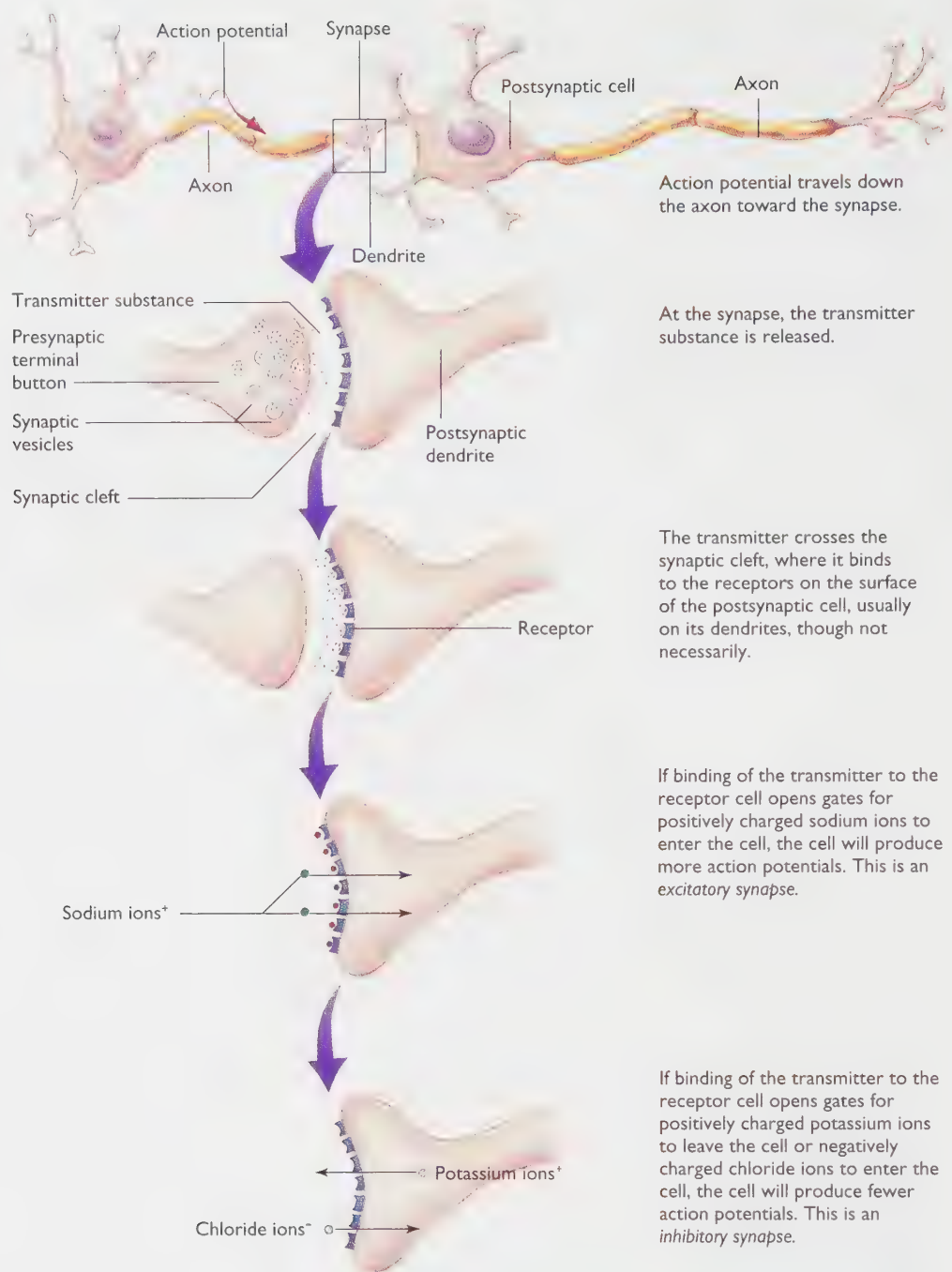


FIGURE 3.22 The complex process of neural communication actually takes only 1–2 milliseconds.

reuptake and therefore prolonging the effects of one transmitter or another.

CONCEPT CHECKS

- Under some conditions, the axons and dendrites of a neuron increase their branching. How will that affect the number of synapses present?
- Norepinephrine is a neurotransmitter that inhibits postsynaptic neurons. If a drug were injected to prevent norepinephrine from attaching to its receptors, what would happen to the postsynaptic neuron? (Check your answers on page 84.)

WHAT IS THE EVIDENCE? Neurons Communicate by Releasing Chemicals

You have just learned that neurons communicate by releasing chemicals at synapses. Perhaps you are perfectly content to take my word for it and go on with something else. Still, it is advisable to pause and contemplate the evidence responsible for an important conclusion.

Today, neuroscientists have a wealth of evidence that neurons release chemicals at synapses. They can work with

radioactively labeled chemicals that enable investigators to trace where chemicals go and what happens when they get there; they also can inject purified chemicals at a synapse and use extremely fine electrodes to measure the response of the postsynaptic neuron. But scientists have known since the 1920s that neurons communicate by releasing chemicals, although back then they had none of the fancy equipment that they have today. Otto Loewi found evidence of chemical transmission using a very simple, clever experiment, as he later described in his autobiography (Loewi, 1960).

Hypothesis If a neuron releases chemicals, an investigator should be able to collect some of those chemicals and transfer them from one animal to another, and thereby get the second animal to do what the first animal had been doing. Loewi had no method of collecting chemicals released within the brain itself, so he worked with axons communicating with the heart muscle. (Much later research confirmed, as Loewi suspected, that the communication between a neuron and a muscle is similar to that between two neurons.)

Method Loewi began by electrically stimulating some axons connected to a frog's heart. These particular axons slowed down the heart rate. As he continued to stimulate those axons, he collected some of the fluid on and around that heart and transferred it to the heart of a second frog.

Results When Loewi transferred the fluid from the first frog's heart, the second frog's heart rate also slowed down (Figure 3.23).

Interpretation Evidently, the stimulated axons had released a chemical that slows heart rate. At least in this case, neurons send messages by releasing chemicals.

This experiment was remarkably clever; indeed, Loewi eventually won a Nobel Prize in physiology for this and related experiments. Even outstanding experiments have limitations, however. In this case, the main limitation was the uncertainty about whether the conclusion applied only to frog hearts or whether it applied to all communication by neurons. Answering *that* question required enormous efforts and much more elaborate equipment. (The answer is that *almost* all communication by neurons depends on the release of chemicals; a few exceptional neurons communicate electrically.)



Neurotransmitters and Behavior: The Example of Parkinson's Disease

Dozens of chemicals serve as neurotransmitters at one brain location or another. It has become increasingly difficult to summarize the behavioral effects of different trans-

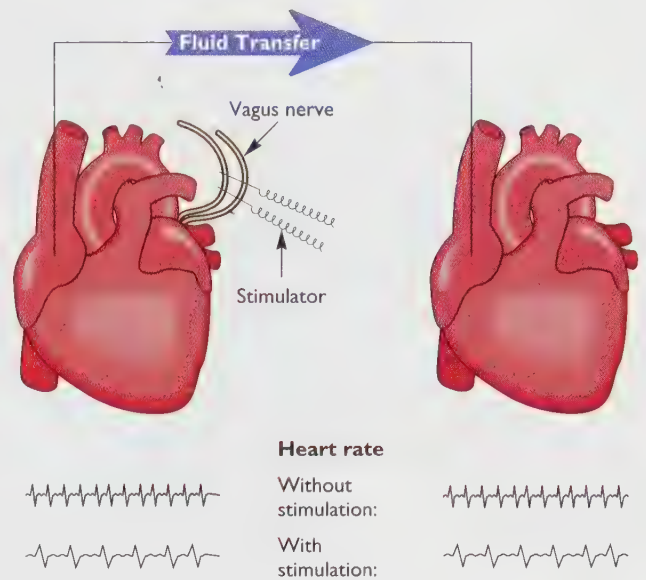


FIGURE 3.23 Otto Loewi demonstrated that axons release chemicals that can affect other cells. Using a frog, he electrically stimulated a set of axons known to decrease the heart rate. Then he collected some fluid from around the heart and transferred it to the surface of another frog's heart. When that heart slowed its beat, Loewi concluded that the axons in the first heart must have released a chemical that slows the heart rate.

mitters, because researchers have discovered that the effects of any neurotransmitter vary among its different kinds of receptors—for example, there are at least 16 receptors for the neurotransmitter glutamate (Westbrook, 1994) and nearly as many for serotonin (Humphrey, Hartig, & Hoyer, 1993). Depending on the receptor, serotonin can affect mood, arousal, motor activity, aggressive behavior, appetite, nausea, and learning. Other transmitters have similarly wide-ranging and miscellaneous effects.

It is difficult to summarize the effects of an overall increase or decrease in a neurotransmitter throughout the brain, although some disorders can be traced to the loss of a particular kind of receptor or the dysfunction of a transmitter in a particular pathway or brain area. One example is Parkinson's disease, a condition that affects about 1% of people over the age of 50. The main symptoms are difficulty in initiating voluntary movement, slowness of movement, tremors, rigidity, and depressed mood, due to a gradual loss of one pathway of neurons that all use the same neurotransmitter, **dopamine** (DOPE-uh-mein), *a chemical that promotes activity levels and facilitates movement* (Figure 3.24). One common treatment is to increase the brain's supply of dopamine. However, dopamine, like many other chemicals, cannot cross directly from the blood into the brain. Instead, physicians often prescribe pills containing L-dopa, which can cross into the brain, which converts L-dopa into dopamine. L-dopa does not prevent the continuing loss of neurons, but it reduces the symptoms of the disease. For people with a mild case of Parkinson's disease, L-dopa can provide added years of nearly normal life; however, it

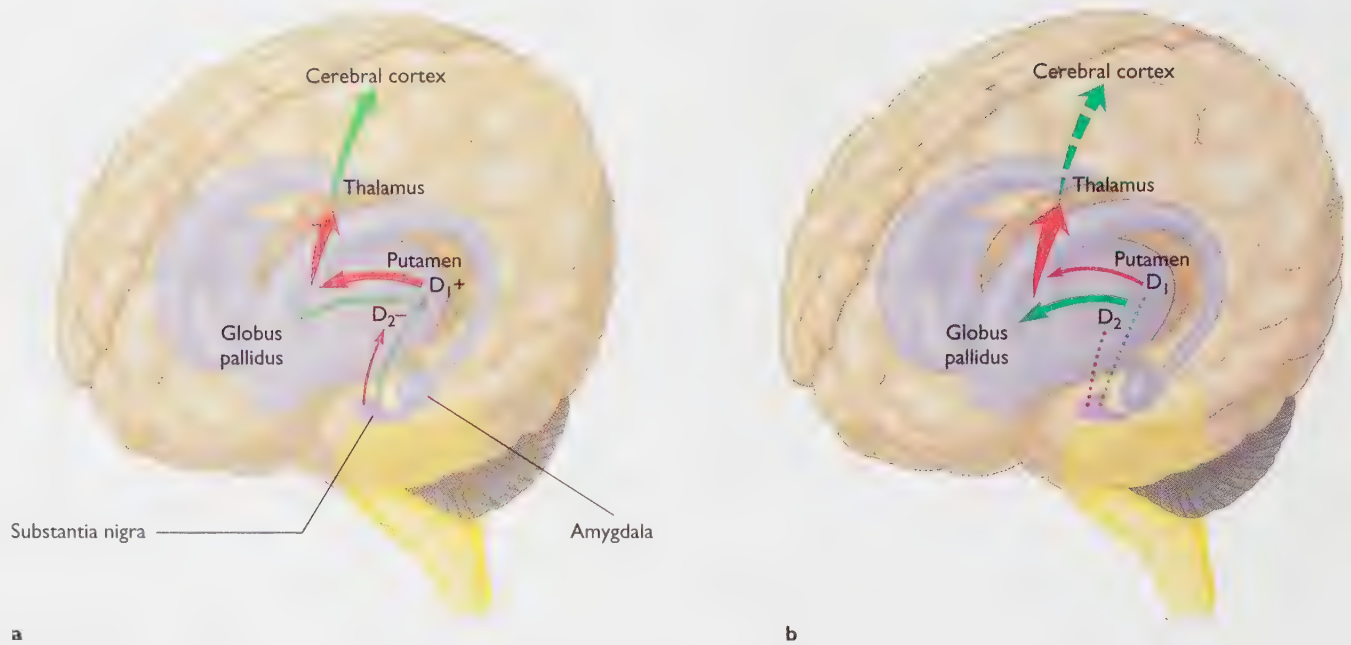


FIGURE 3.24 With Parkinson's disease, axons from the substantia nigra gradually die. (a) Normal brain. (b) Brain of person with Parkinson's disease. Green = excitatory path; red = inhibitory

also produces side effects, including nausea, and the benefits of L-dopa become smaller as the underlying condition deteriorates.

Some additional treatments currently in the experimental stage may become useful in the future. One ap-

proach is to implant dopamine-containing neurons from the brain of a human fetus, but this approach faces not only the ethical difficulties of dealing with aborted fetuses, but also serious practical and scientific problems (Kupsch, Oertel, Earl, & Sautter, 1995). Another experimental approach is to inject certain chemicals that promote the survival and growth of the remaining neurons in the damaged brain area (Gash et al., 1996; Kolb, Cote, Ribeiro-da-Silva, & Cuellar, 1997). That approach appears promising but needs much additional research.

CONCEPT CHECK

9. People suffering from certain disorders are given haloperidol, a drug that blocks activity at dopamine synapses. How would haloperidol affect a person suffering from Parkinson's disease? (Check your answer on page 84.)



Former boxing champion Muhammad Ali developed symptoms of Parkinson's disease, presumably because of blows to his head.

THE MESSAGE *From Neurons to Behavior*

Here is an imperfect analogy for the nervous system: At U.S. high-school or college football games, one section of the stands is sometimes set aside for fans holding large cards. When their leader calls out a signal, such as "16," each of them holds up the appropriate card. If you were one of the fans, your card #16 might be all red, providing no clue about what the overall message might be. But the pat-